# Priority Queues

Two kinds of priority queues:

- Min priority queue.
- Max priority queue.

# Min Priority Queue

- Collection of elements.
- Each element has a priority or key.
- Supports following operations:
  - empty
  - size
  - insert an element into the priority queue (push)
  - get element with min priority (top)
  - remove element with min priority (pop)

# Max Priority Queue

- Collection of elements.
- Each element has a priority or key.
- Supports following operations:
  - empty
  - size
  - insert an element into the priority queue (push)
  - get element with max priority (top)
  - remove element with max priority (pop)

# Complexity Of Operations

Use a heap or a leftist tree (both are defined later).

empty, size, and top => O(1) time

insert (push) and remove (pop) => O(log n) time where n is the size of the priority queue

# Applications

Sorting

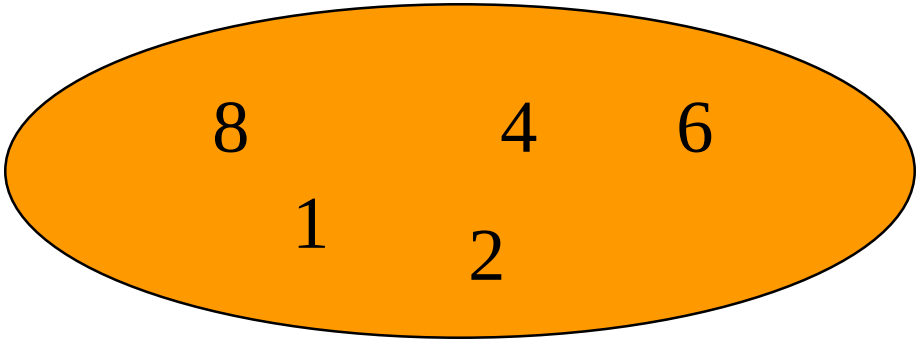
- use element key as priority
- insert elements to be sorted into a priority queue
- remove/pop elements in priority order
  - if a min priority queue is used, elements are extracted in ascending order of priority (or key)
  - if a max priority queue is used, elements are extracted in descending order of priority (or key)

# Sorting Example

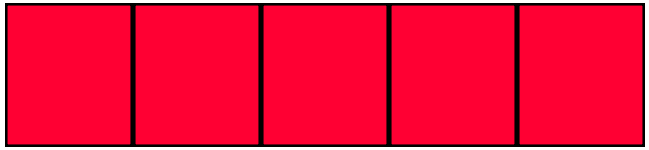
Sort five elements whose keys are 6, 8, 2, 4, 1 using a max priority queue.

- Insert the five elements into a max priority queue.
- Do five remove max operations placing removed elements into the sorted array from right to left.

# After Inserting Into Max Priority Queue

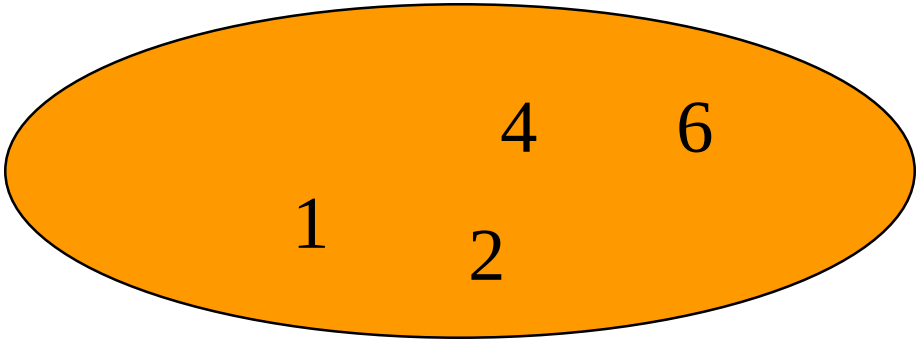



Max Priority Queue

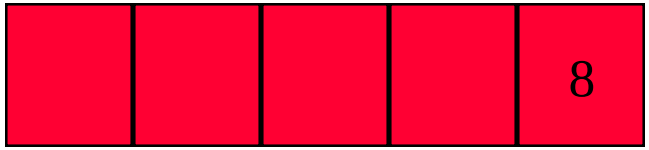


Sorted Array

# After First Remove Max Operation

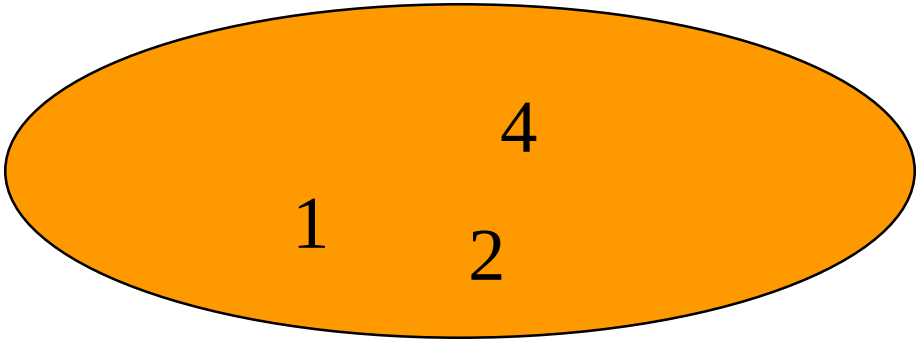



Max Priority Queue

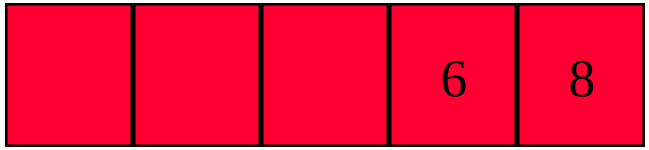



Sorted Array

# After Second Remove Max Operation

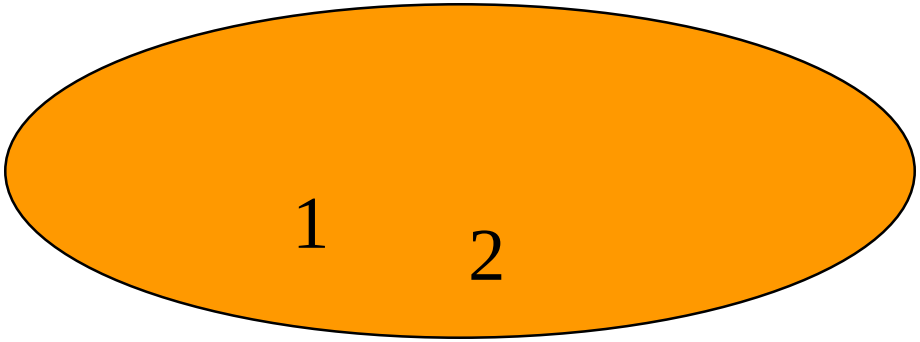



Max Priority Queue

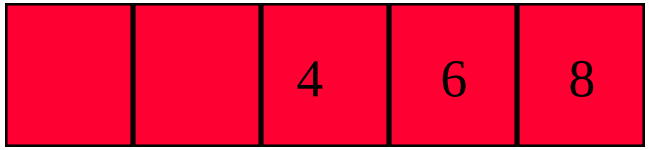



Sorted Array

# After Third Remove Max Operation

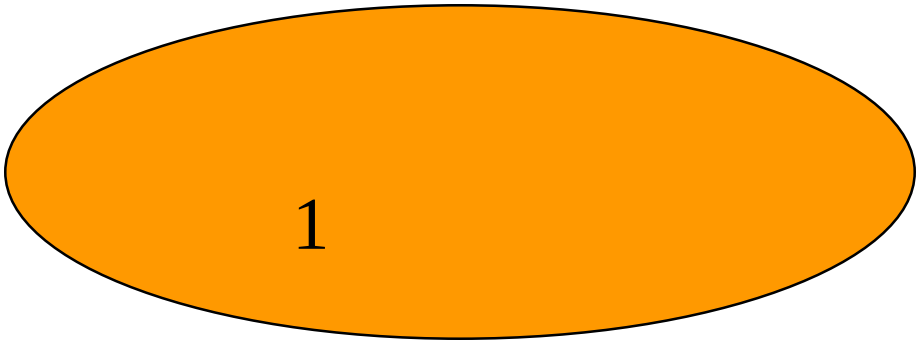



Max Priority Queue

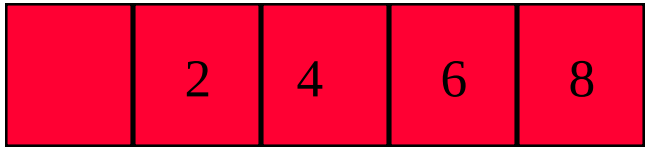



Sorted Array

# After Fourth Remove Max Operation

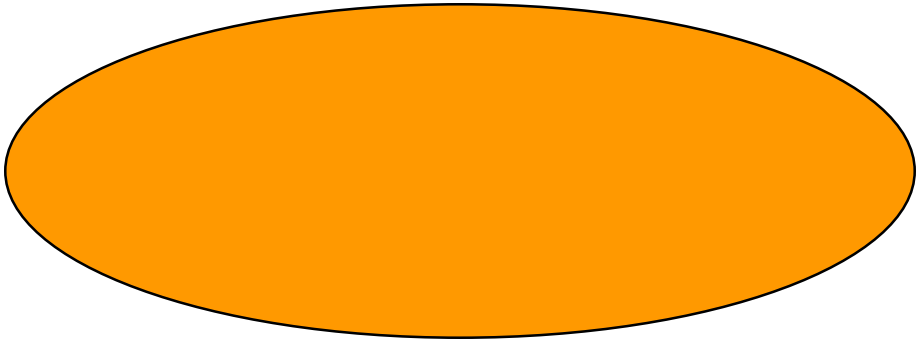



Max Priority Queue




Sorted Array

# After Fifth Remove Max Operation



Max Priority Queue

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 4 | 6 | 8 |

Sorted Array

# Complexity Of Sorting

Sort n elements.

- n insert operations => O(n log n) time.
- n remove max operations => O(n log n) time.
- total time is O(n log n).
- compare with $O(n^2)$ for insertion sort of Lecture 1.

# Heap Sort

Uses a max priority queue that is implemented as a heap.

Initial insert operations are replaced by a heap initialization step that takes O(n) time.

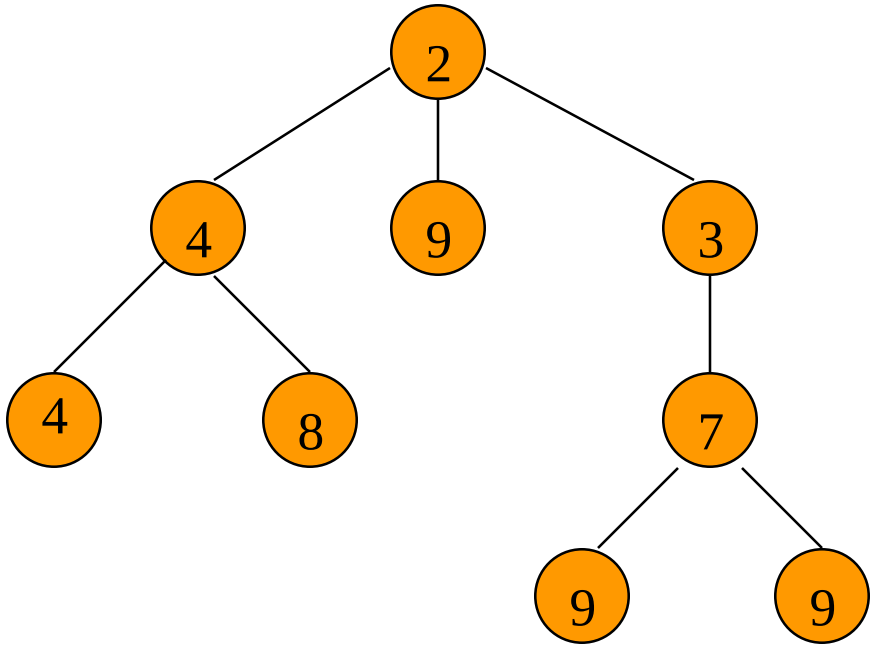
# Min Tree Definition

Each tree node has a value.

Value in any node is the minimum value in the subtree for which that node is the root.

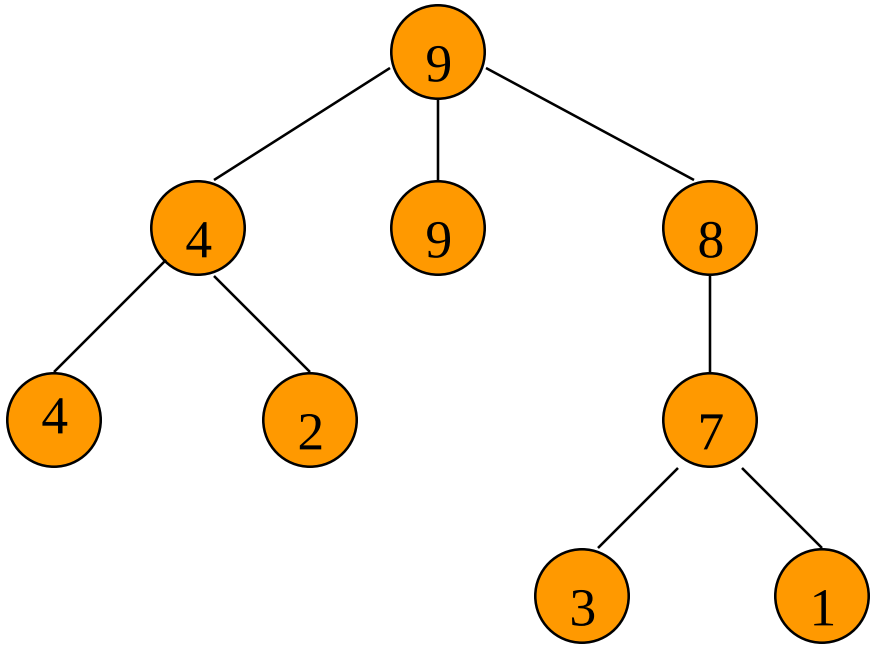
Equivalently, no descendent has a smaller value.

# Min Tree Example




Root has minimum element.

# Max Tree Example

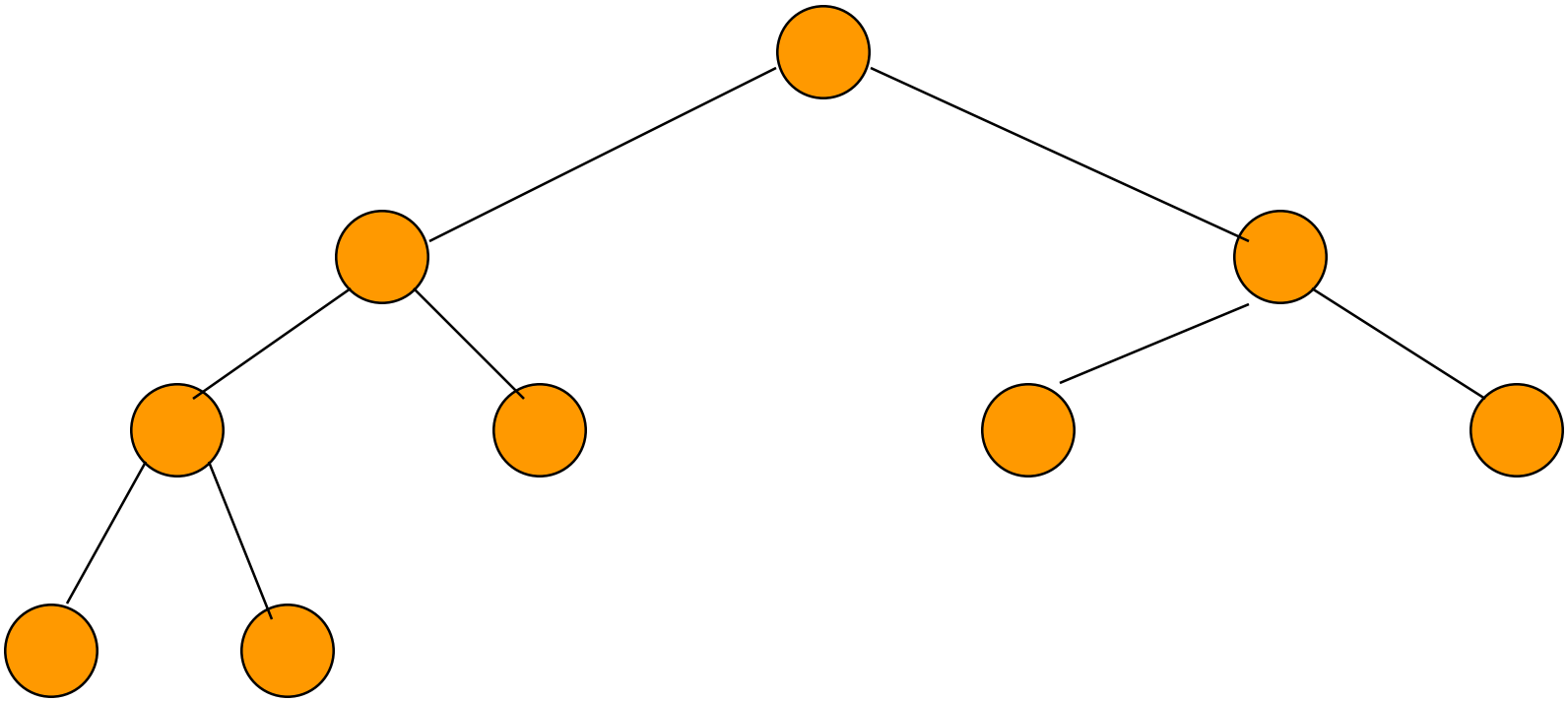



Root has maximum element.

# Min Heap Definition

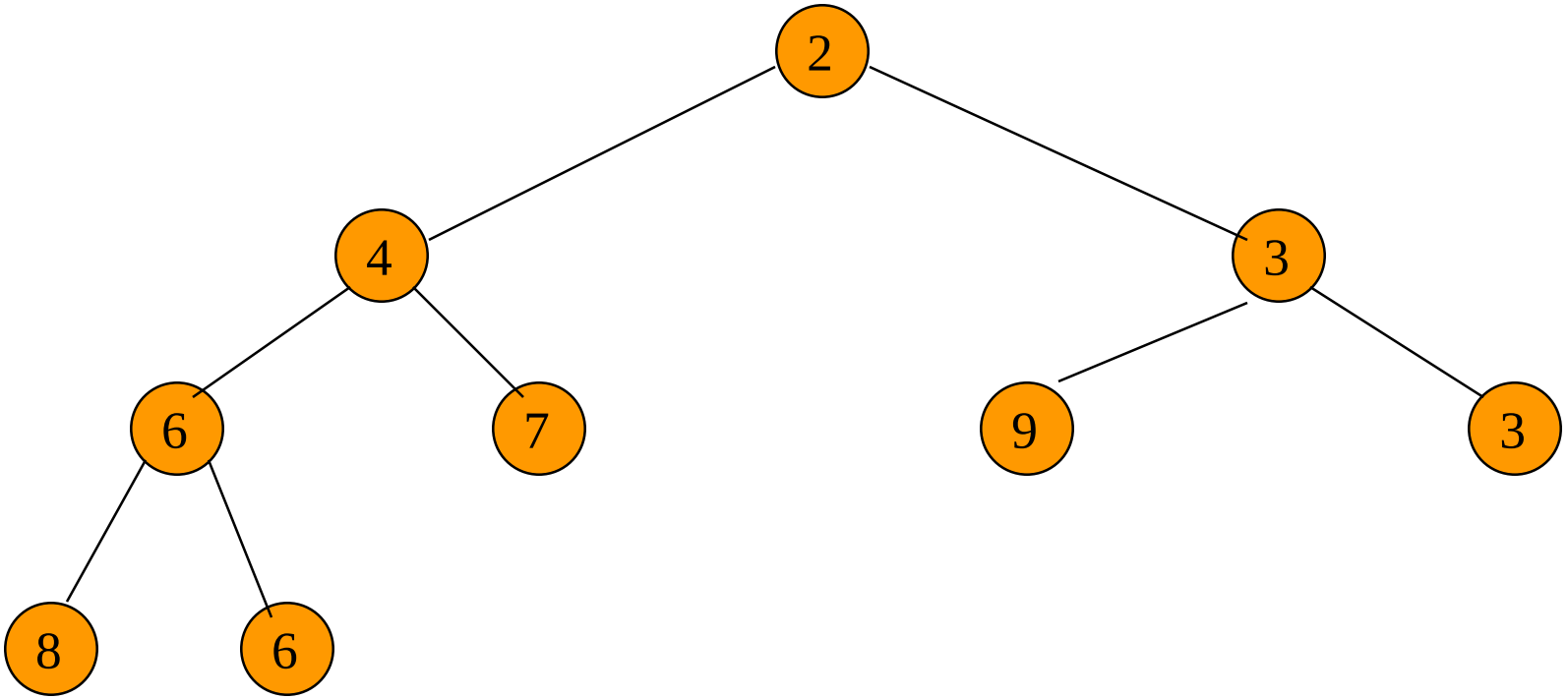
- complete binary tree
- min tree

# Min Heap With 9 Nodes



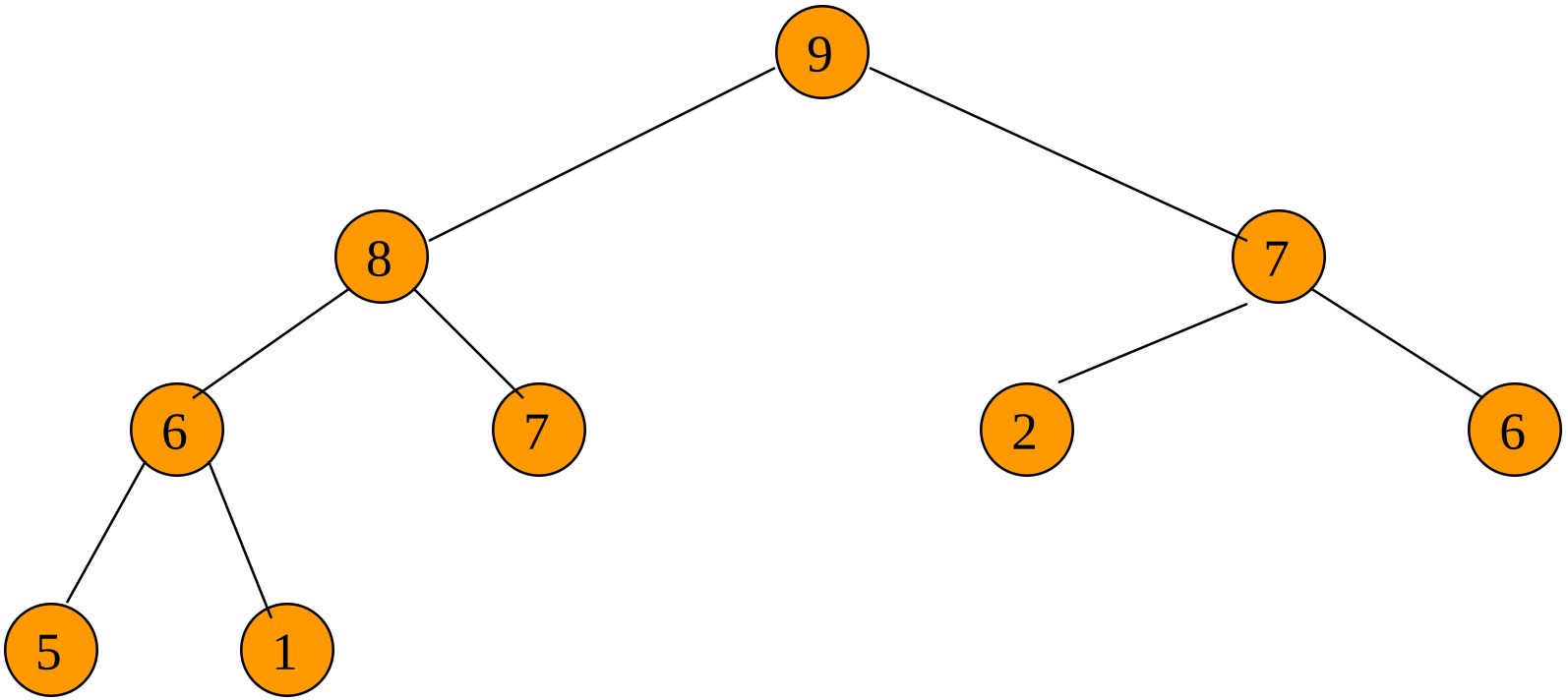
Complete binary tree with 9 nodes.

# Min Heap With 9 Nodes




Complete binary tree with 9 nodes that is also a min tree.

# Max Heap With 9 Nodes

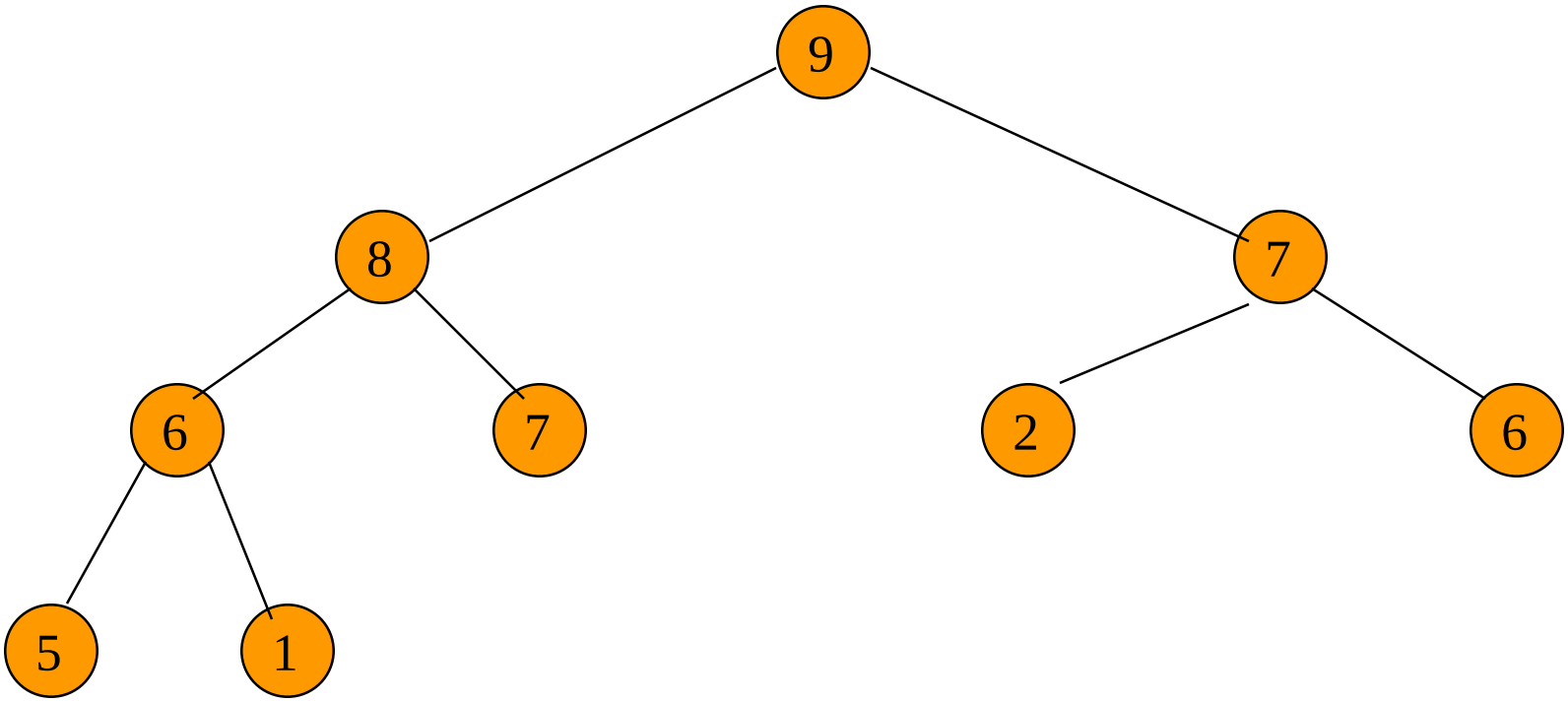



Complete binary tree with 9 nodes that is also a max tree.

# Heap Height

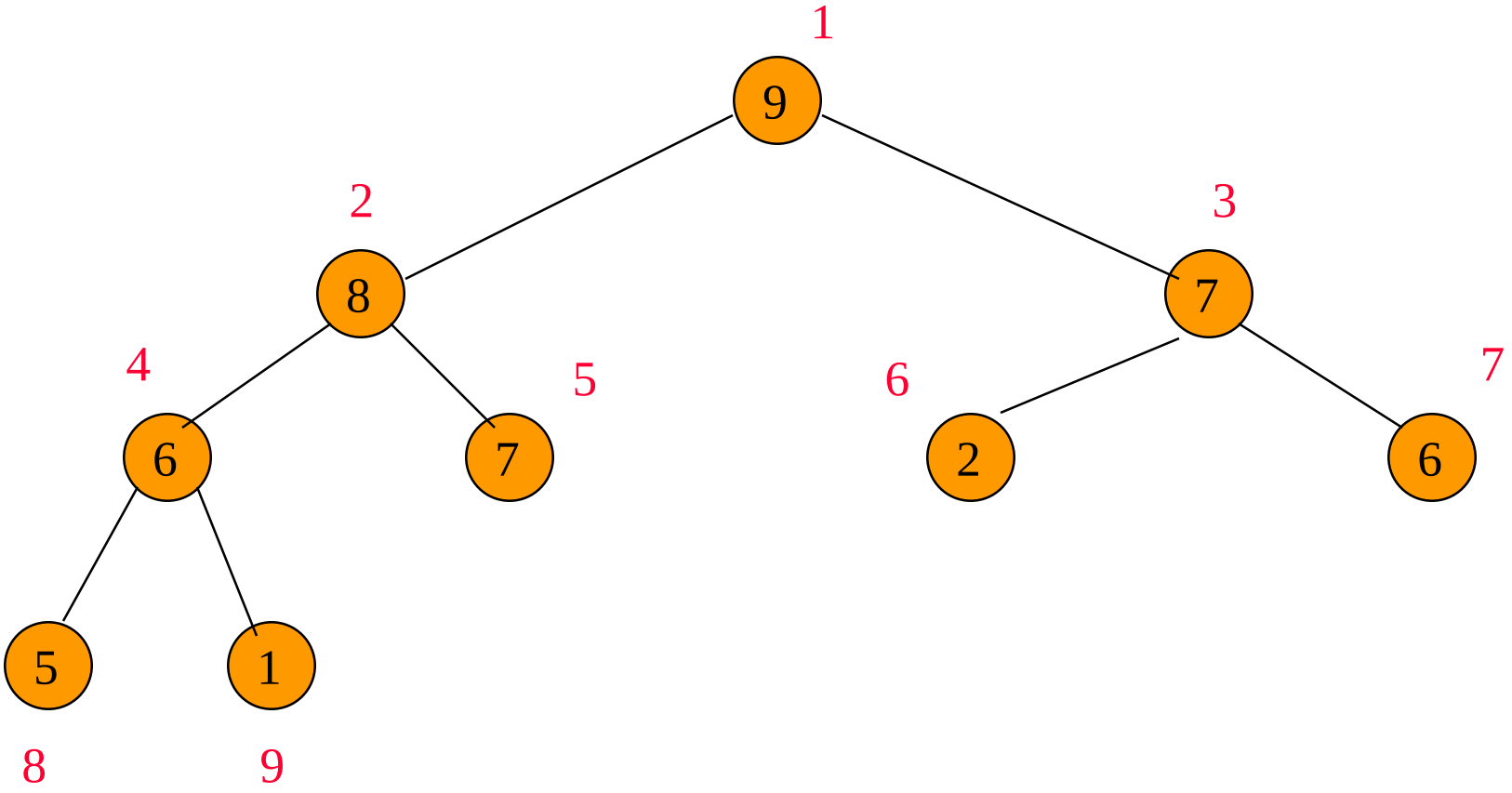
Since a heap is a complete binary tree, the height of an $n$ node heap is $\log_2 (n+1)$.

# A Heap Is Efficiently Represented As An Array

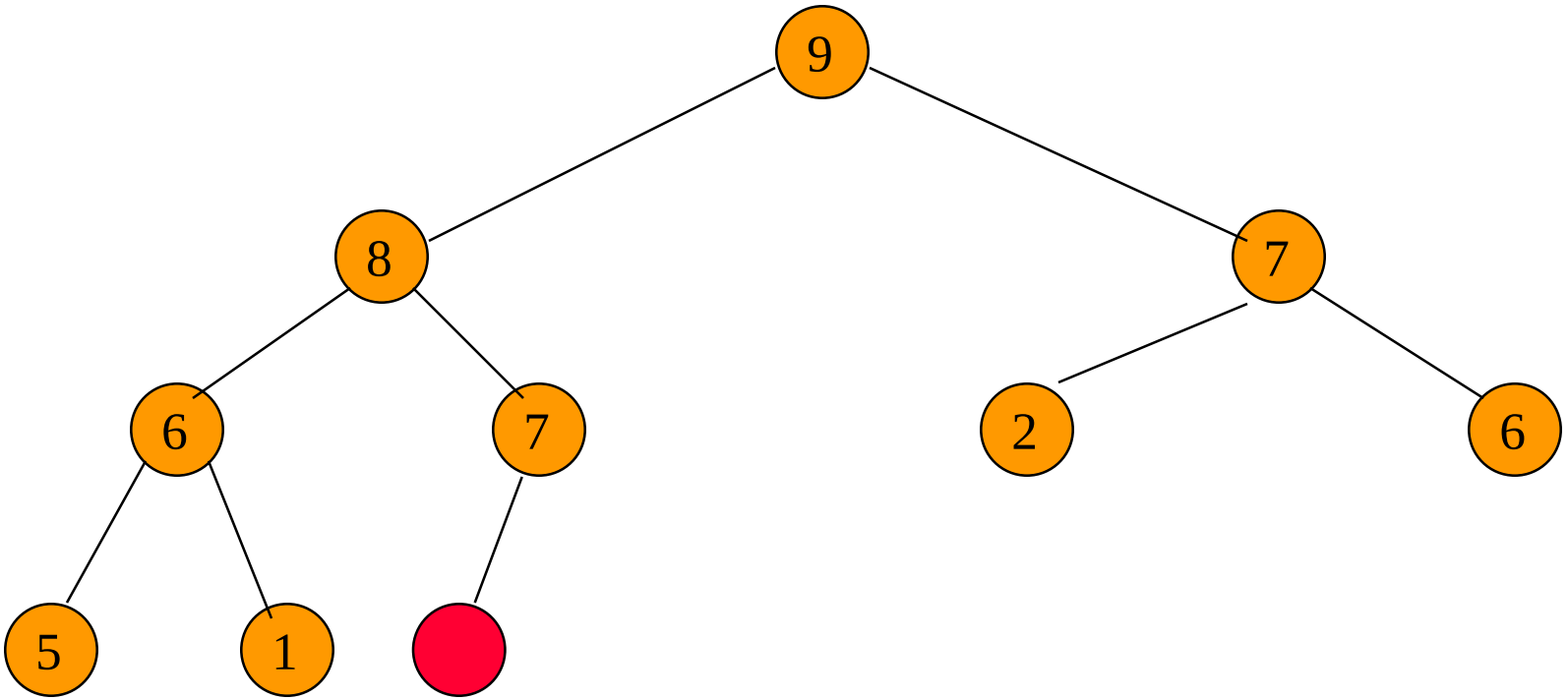



| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 9 | 8 | 7 | 6 | 7 | 2 | 6 | 5 | 1 | | | | | | |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | | | | | |

# Moving Up And Down A Heap

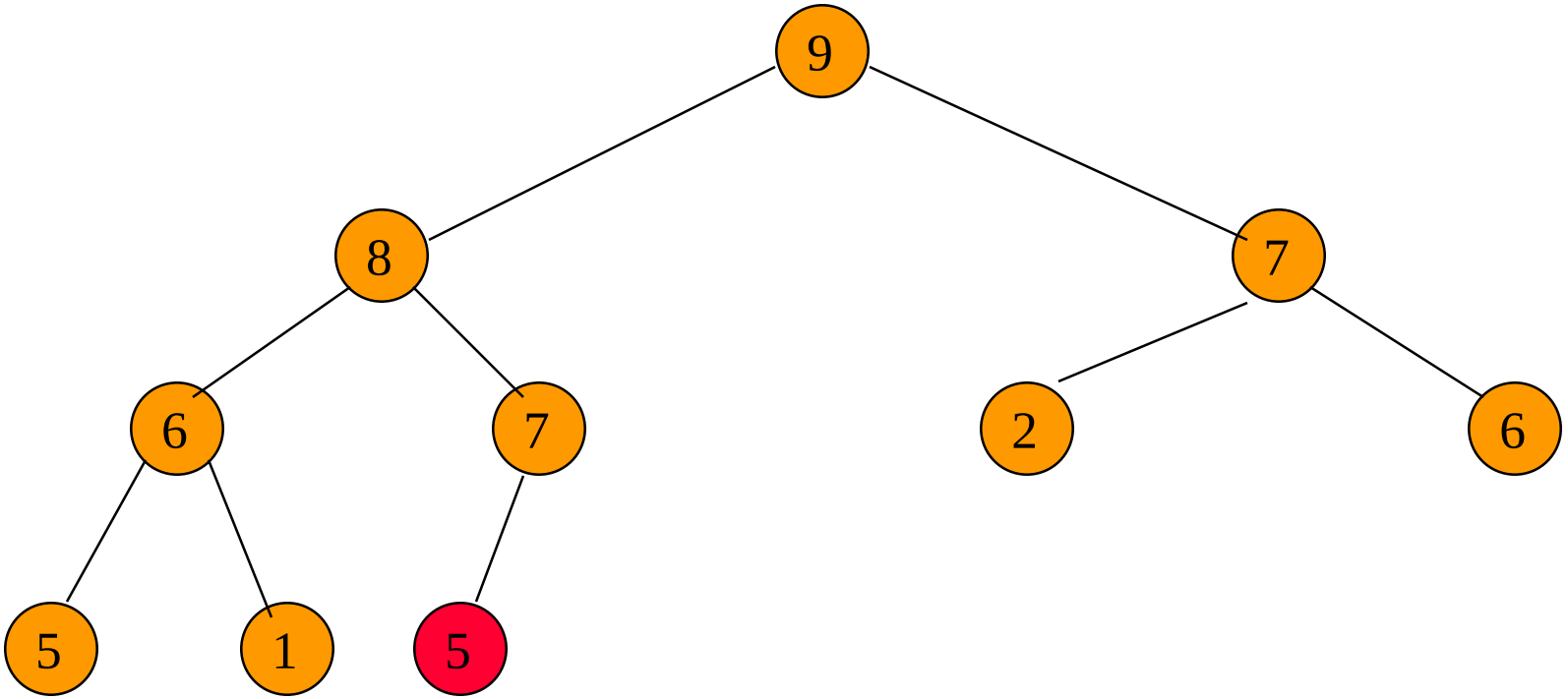


# Inserting An Element Into A Max Heap



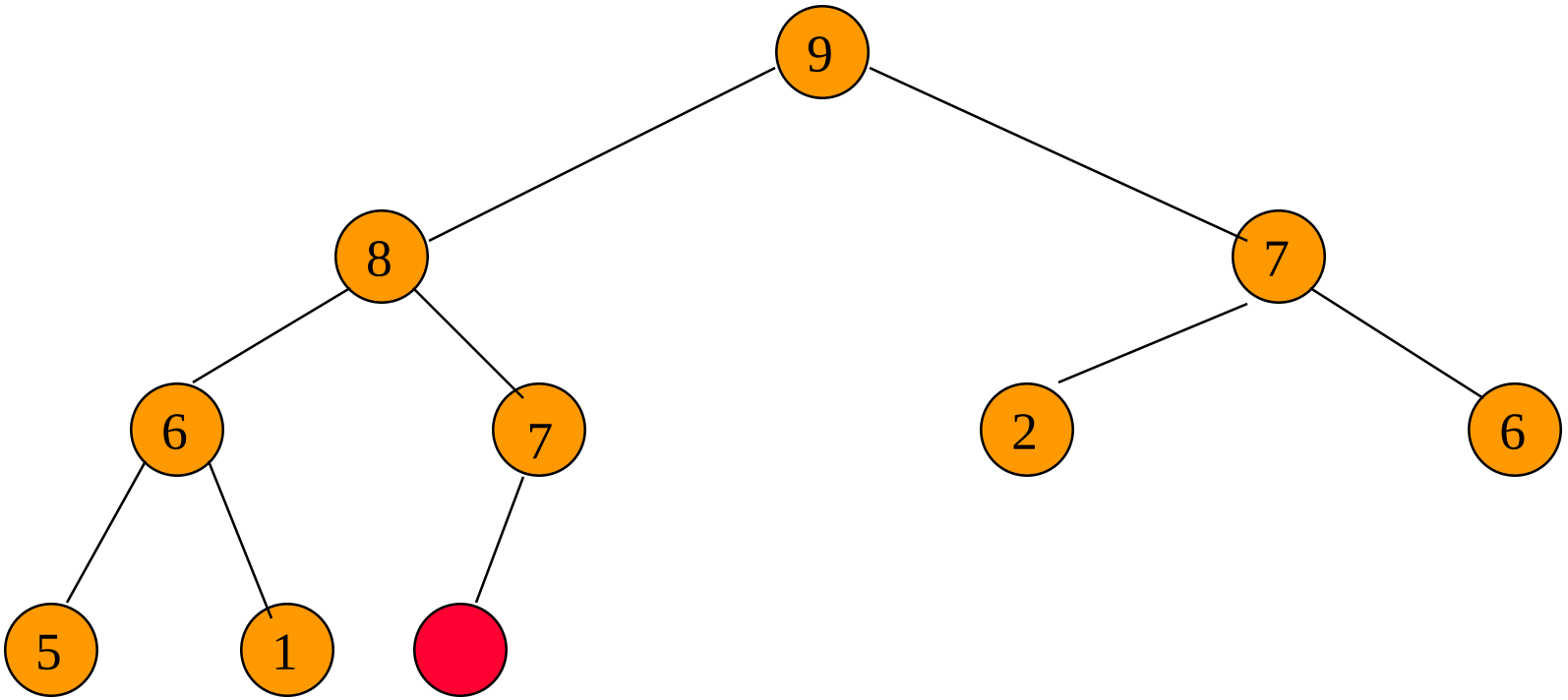

Complete binary tree with 10 nodes.

# Inserting An Element Into A Max Heap



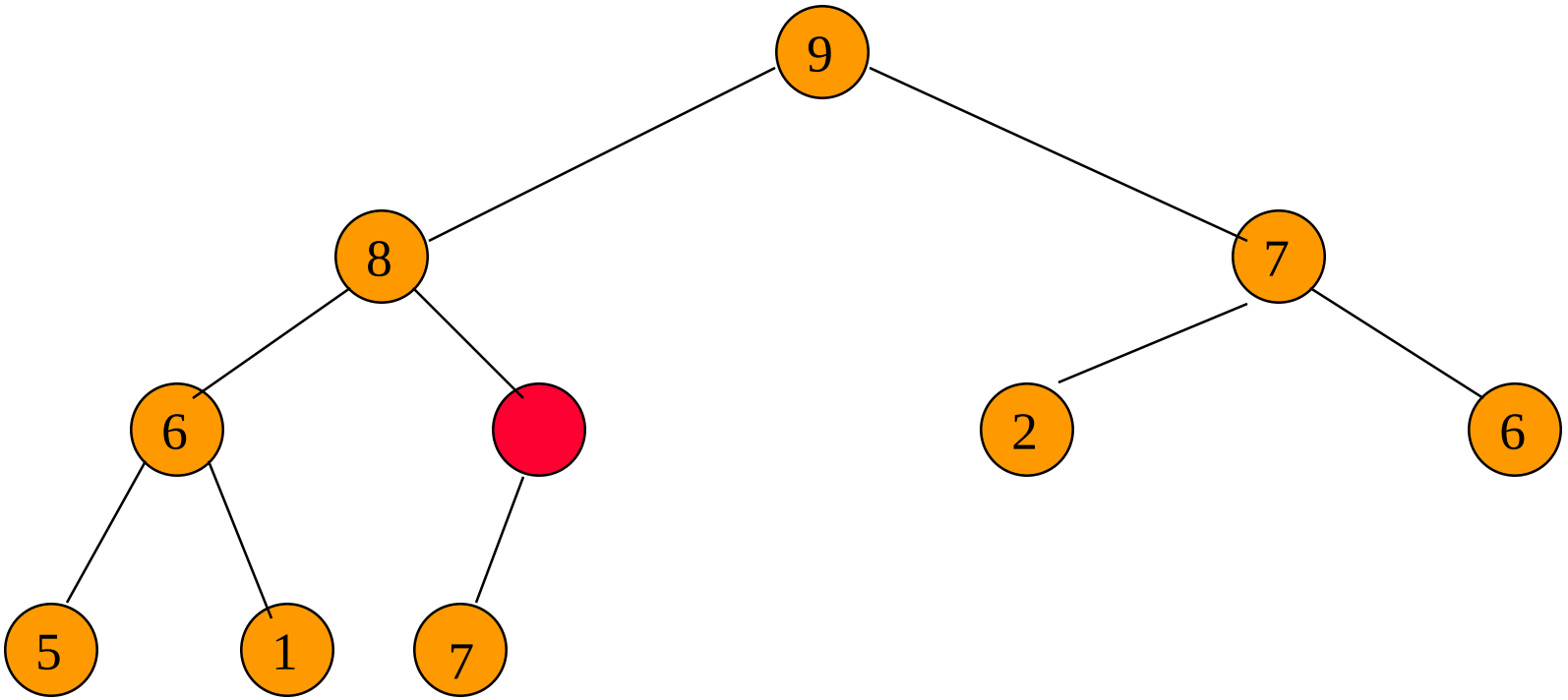

New element is 5.

# Inserting An Element Into A Max Heap



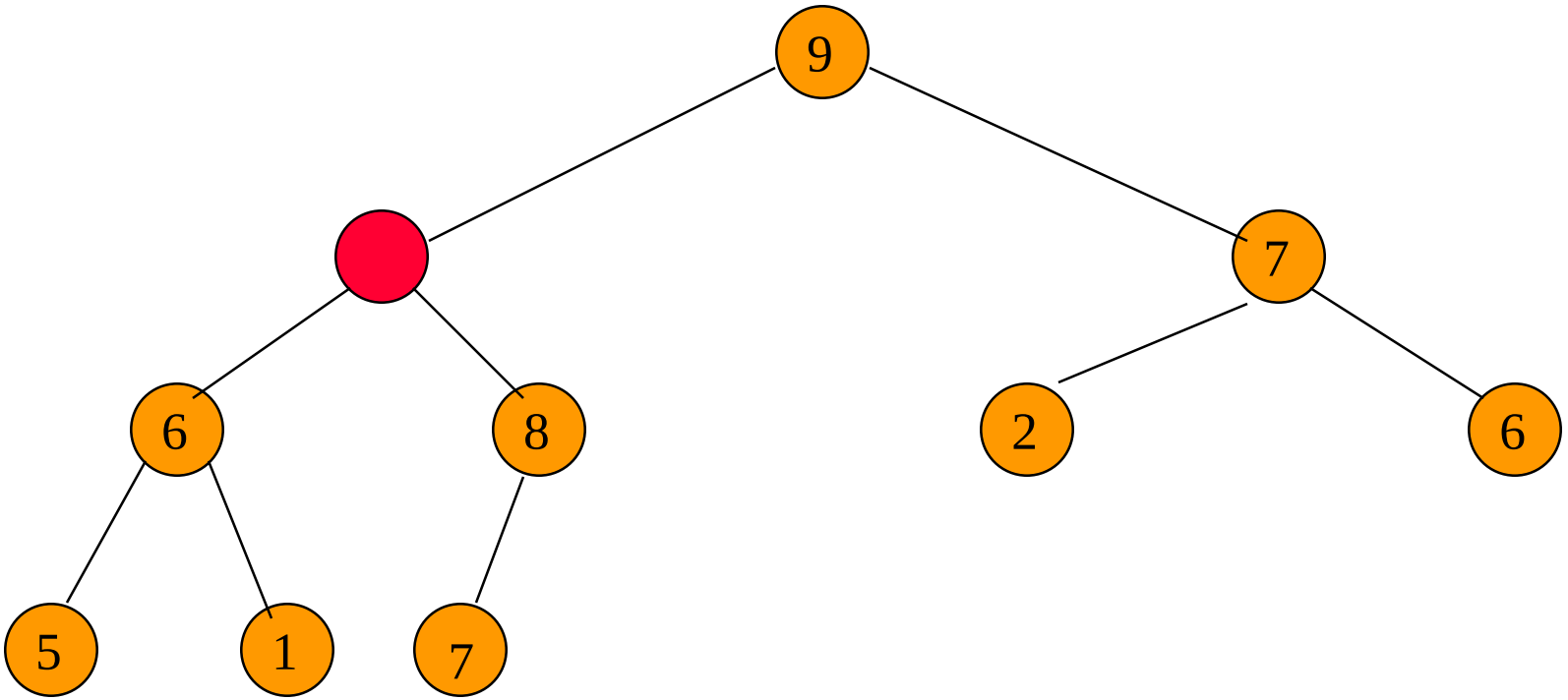

New element is 20.

# Inserting An Element Into A Max Heap



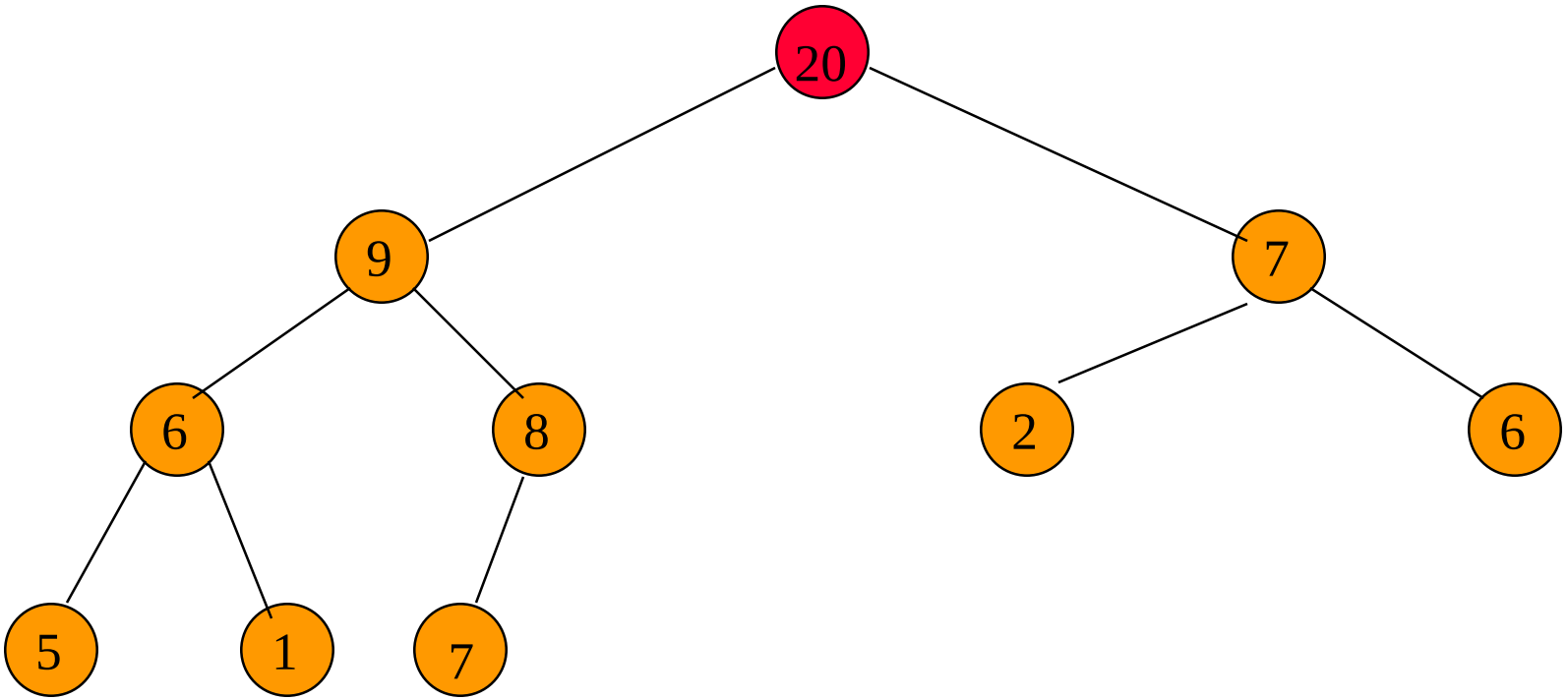

New element is 20.

# Inserting An Element Into A Max Heap



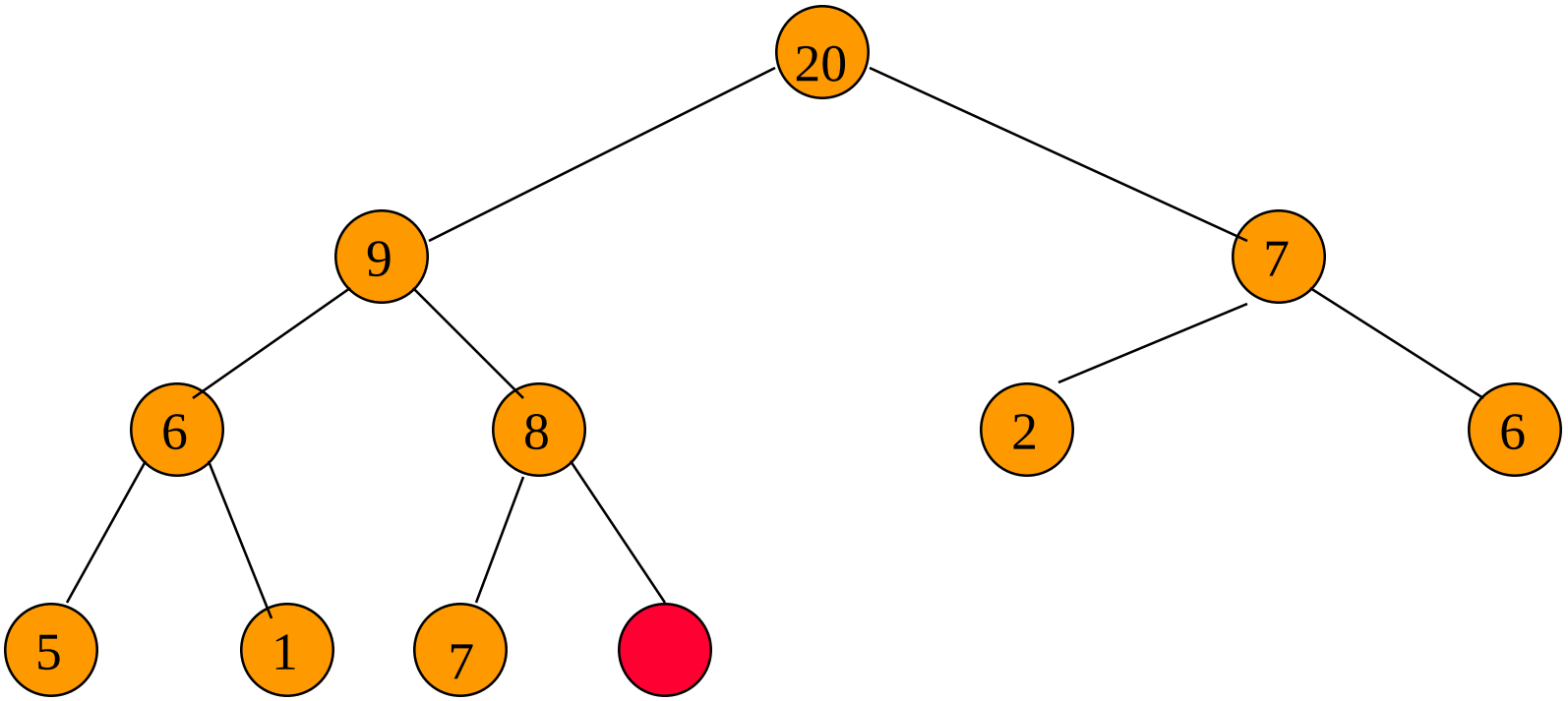

New element is 20.

# Inserting An Element Into A Max Heap



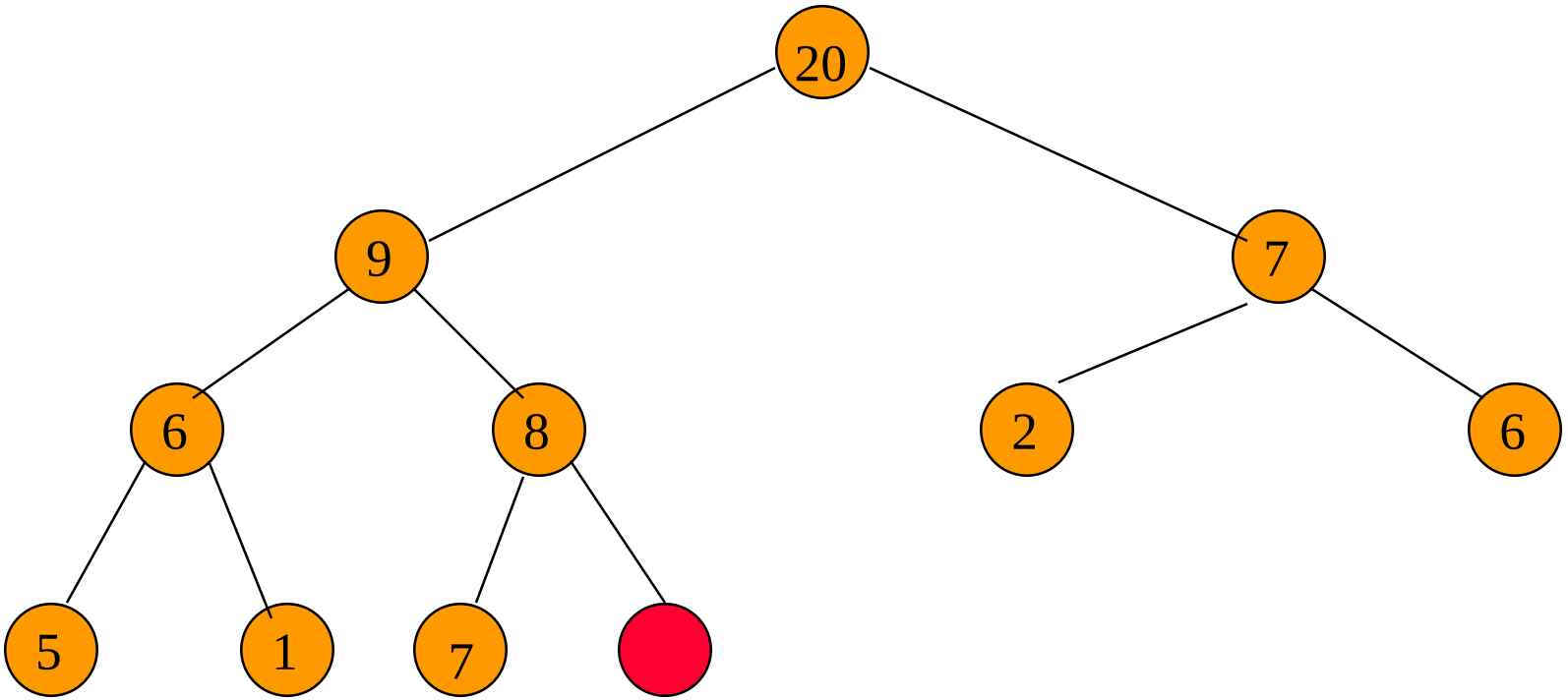

New element is 20.

# Inserting An Element Into A Max Heap



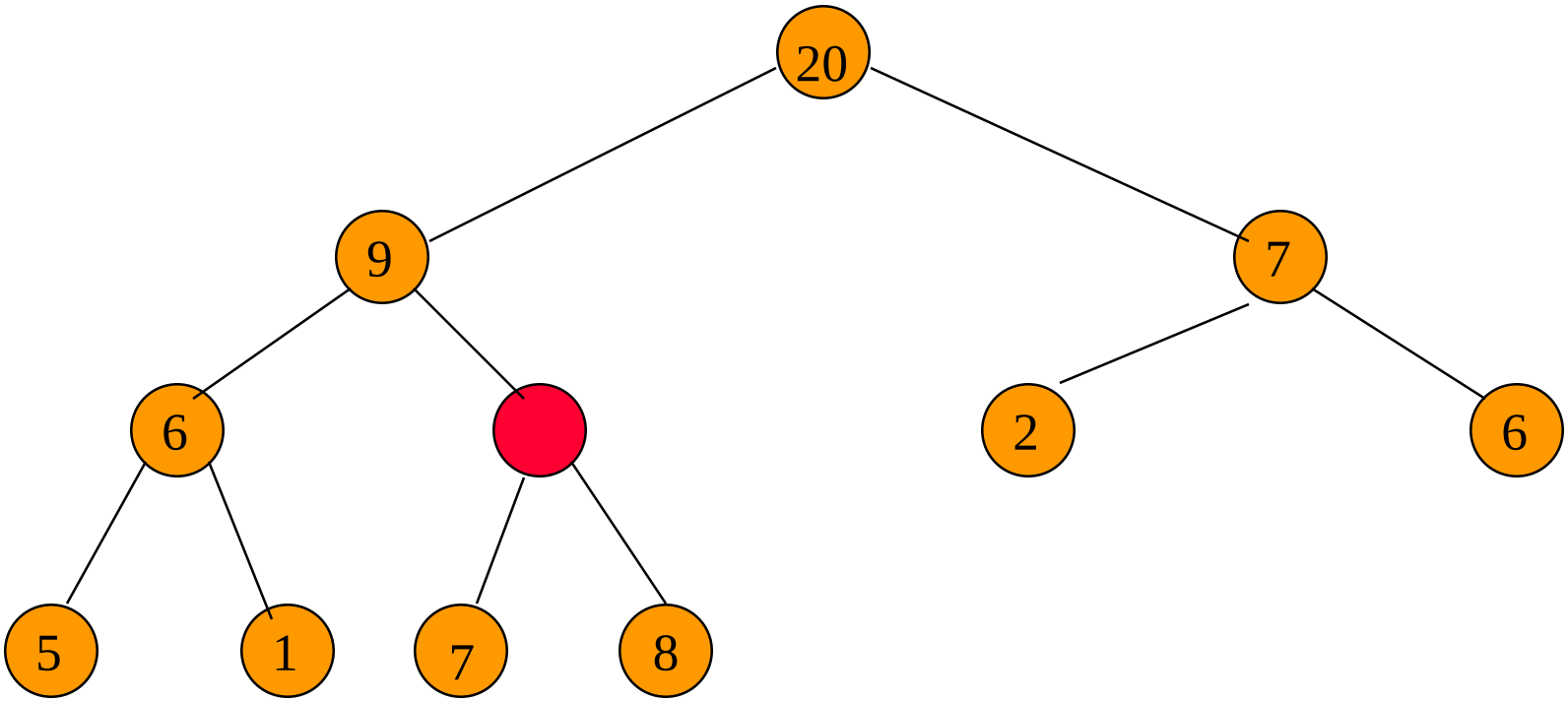

Complete binary tree with 11 nodes.

# Inserting An Element Into A Max Heap



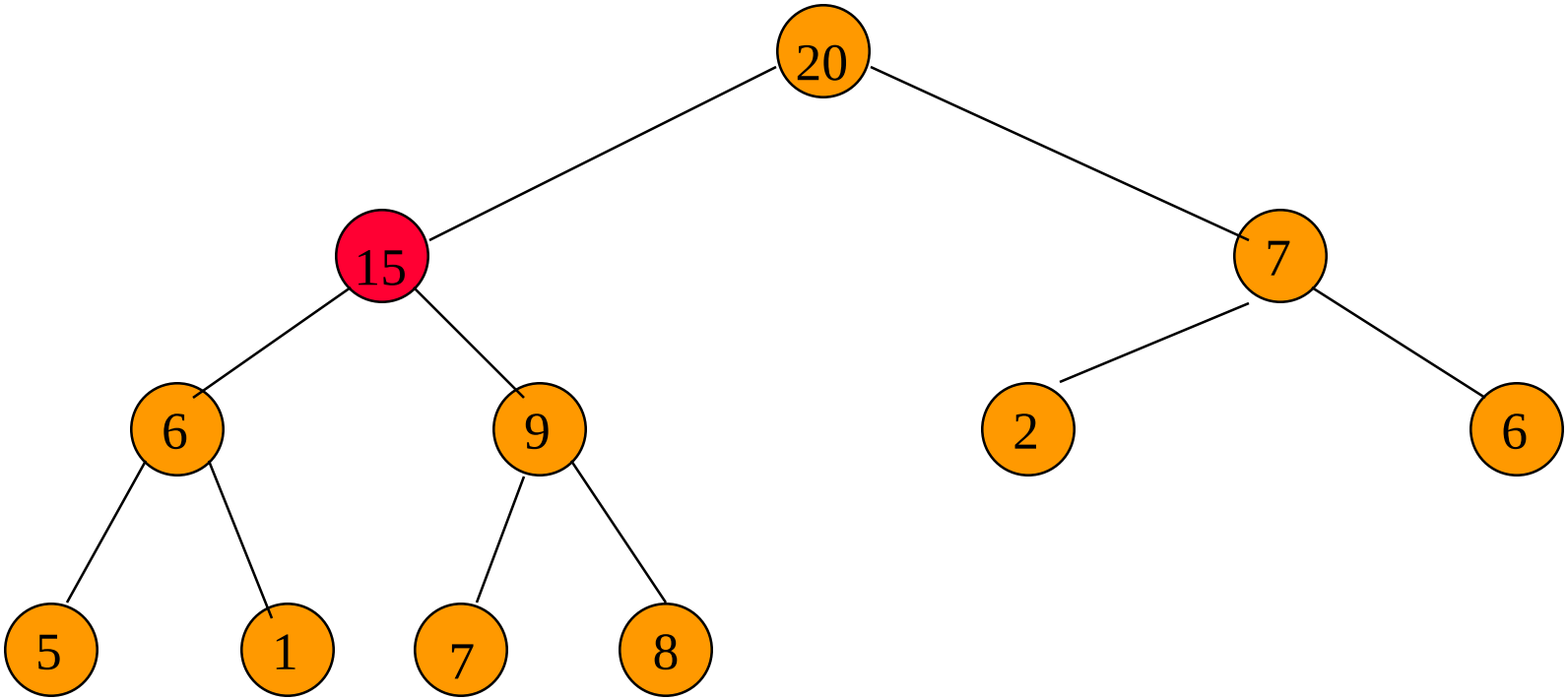

New element is 15.

# Inserting An Element Into A Max Heap



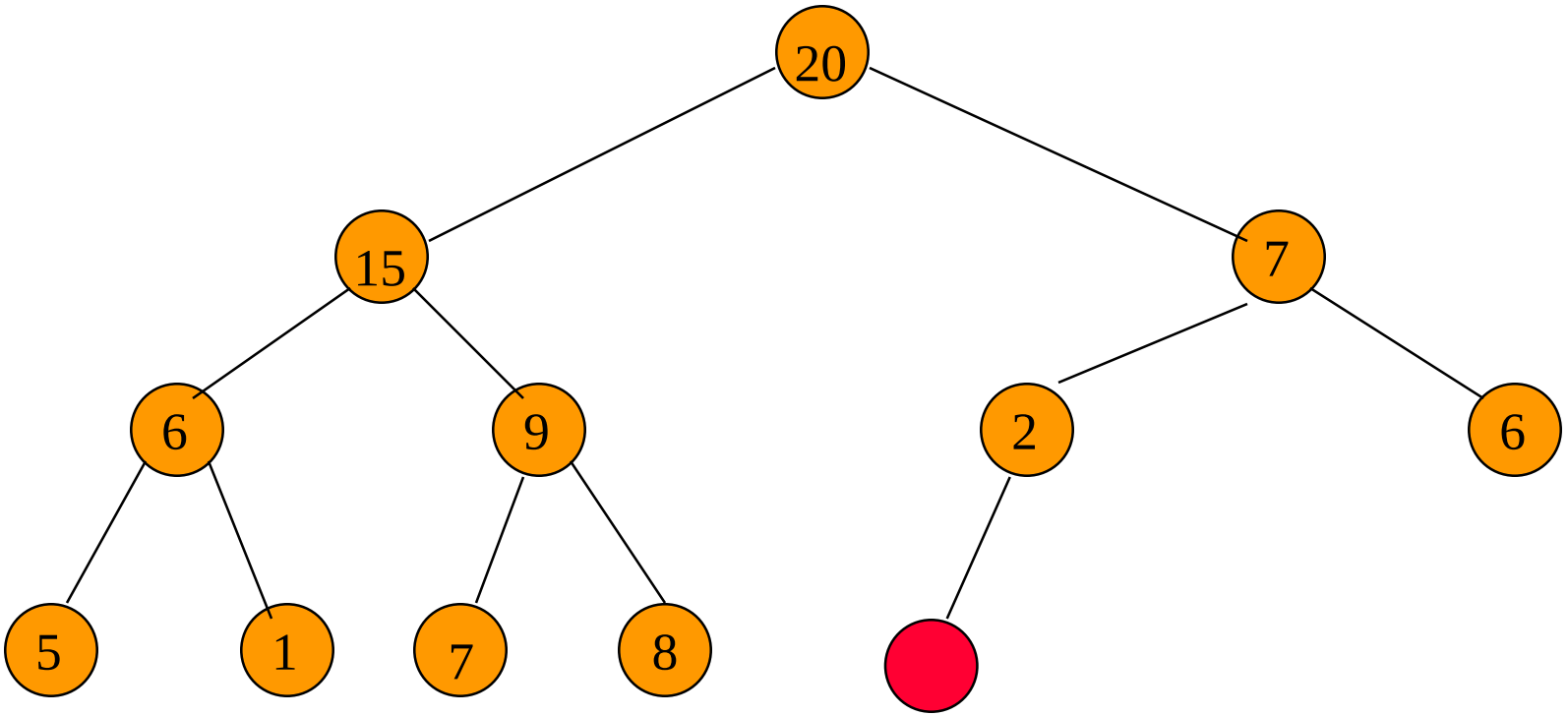

New element is 15.

# Inserting An Element Into A Max Heap



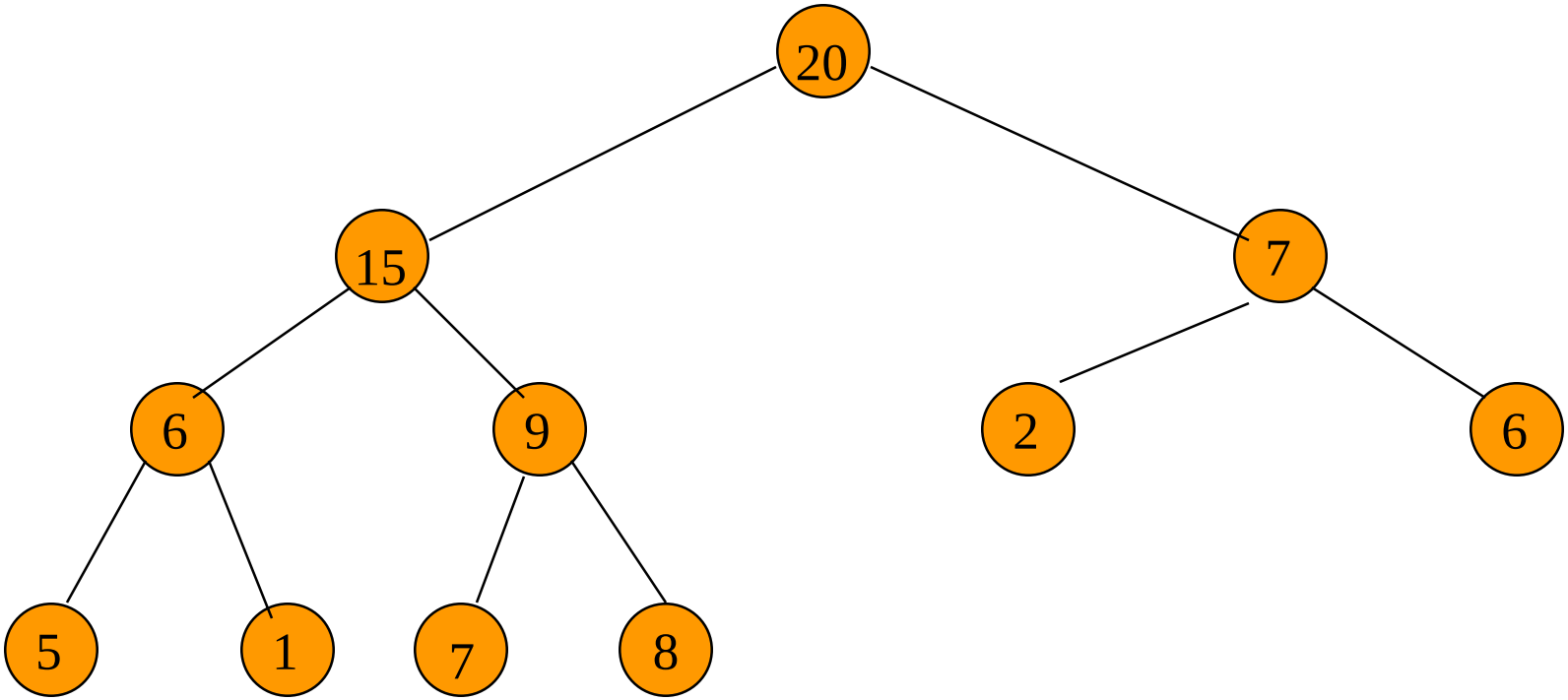

New element is 15.

# Complexity Of Insert



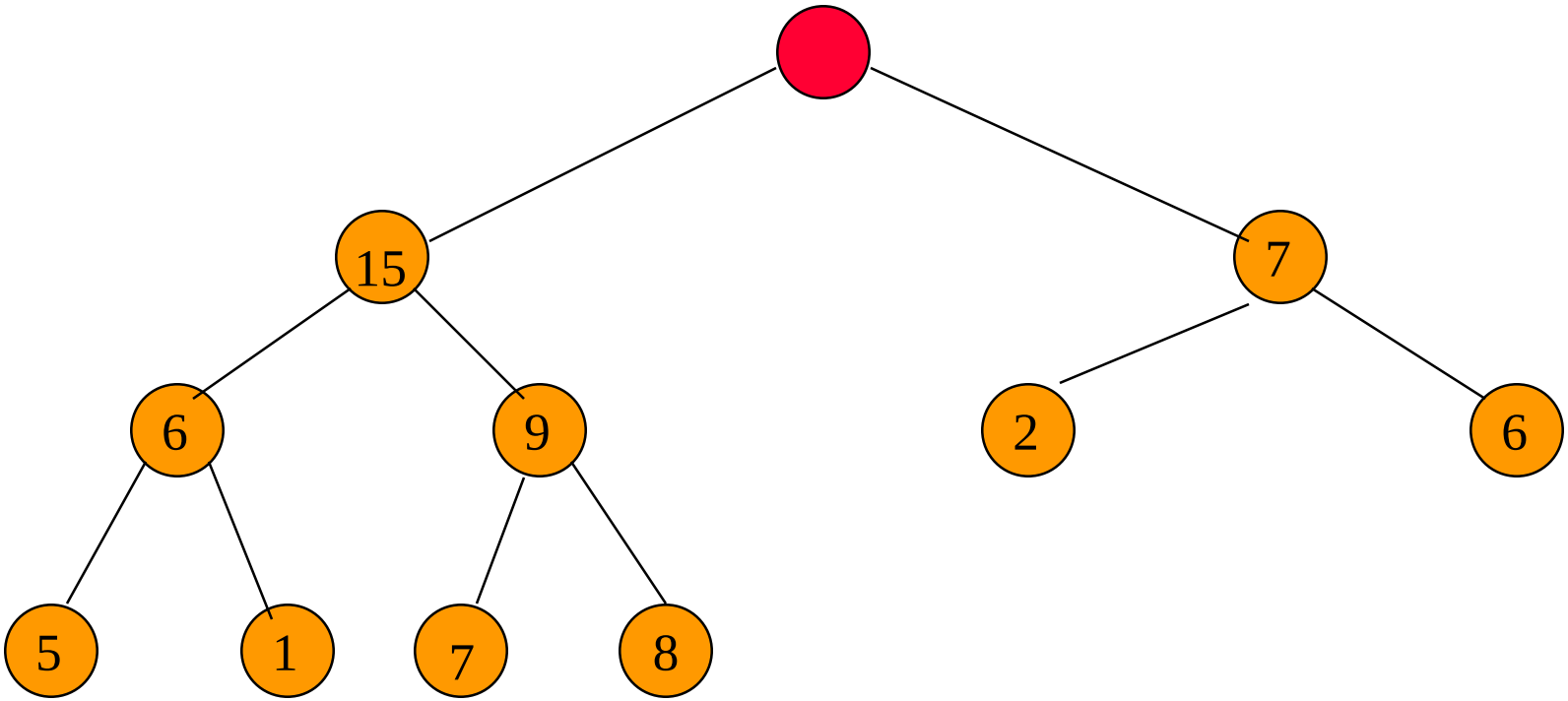

Complexity is O(log n), where n is heap size.

# Removing The Max Element



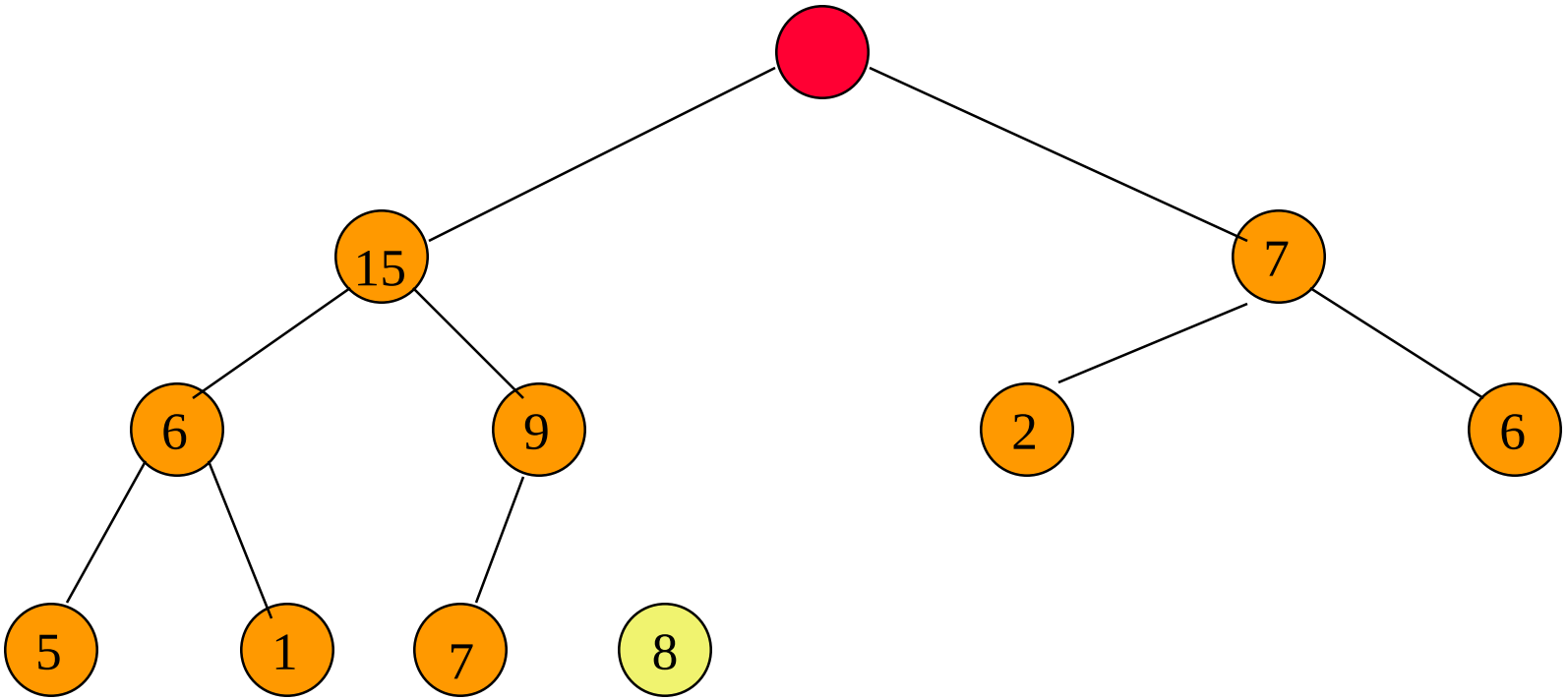

Max element is in the root.

# Removing The Max Element




After max element is removed.

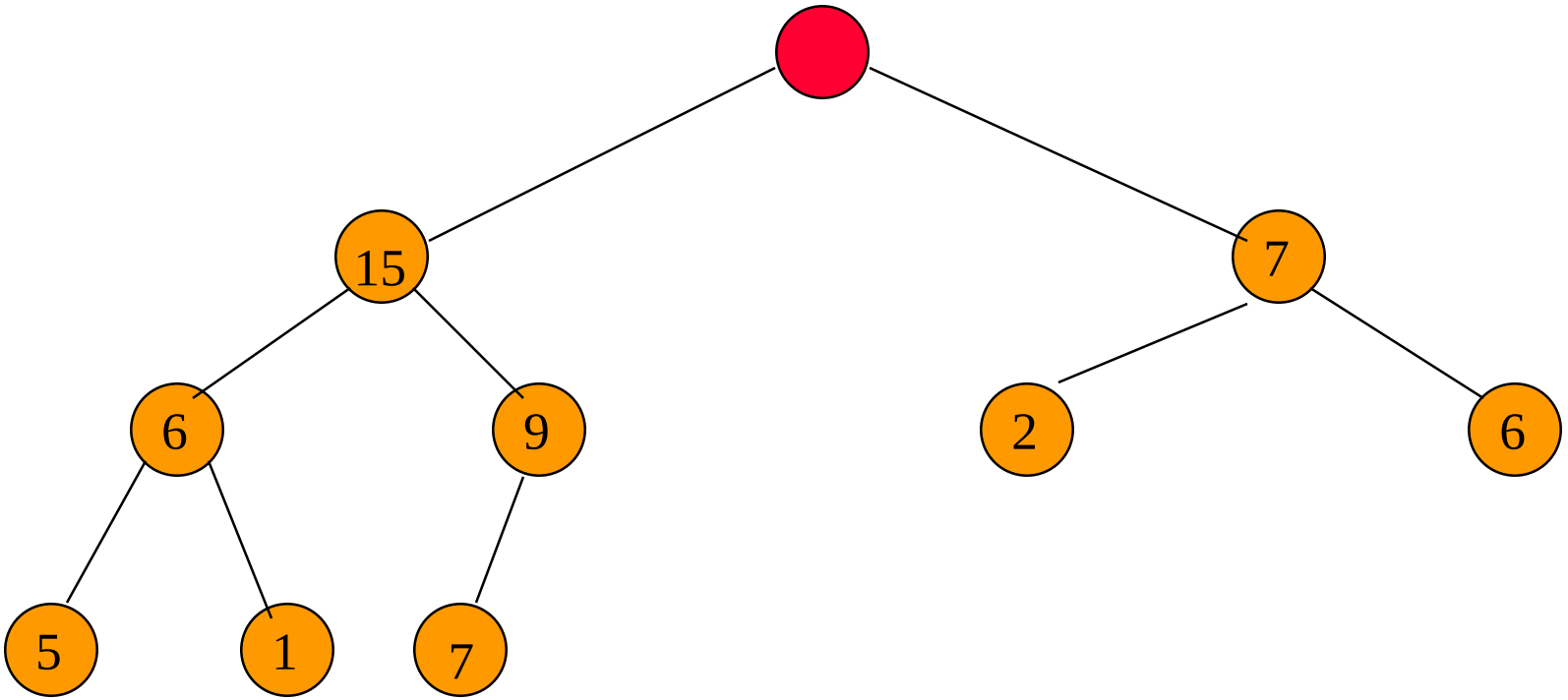
# Removing The Max Element




Heap with 10 nodes.

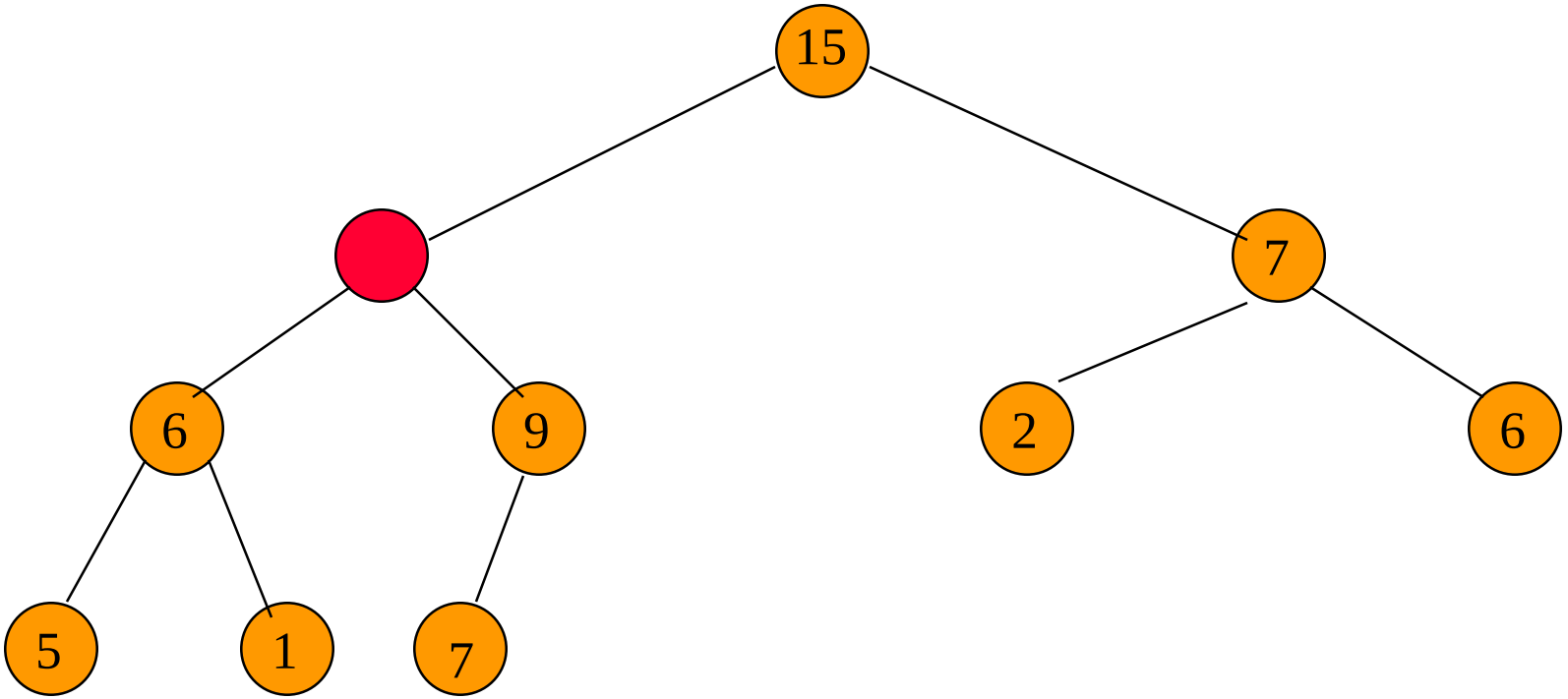
Reinsert 8 into the heap.

# Removing The Max Element



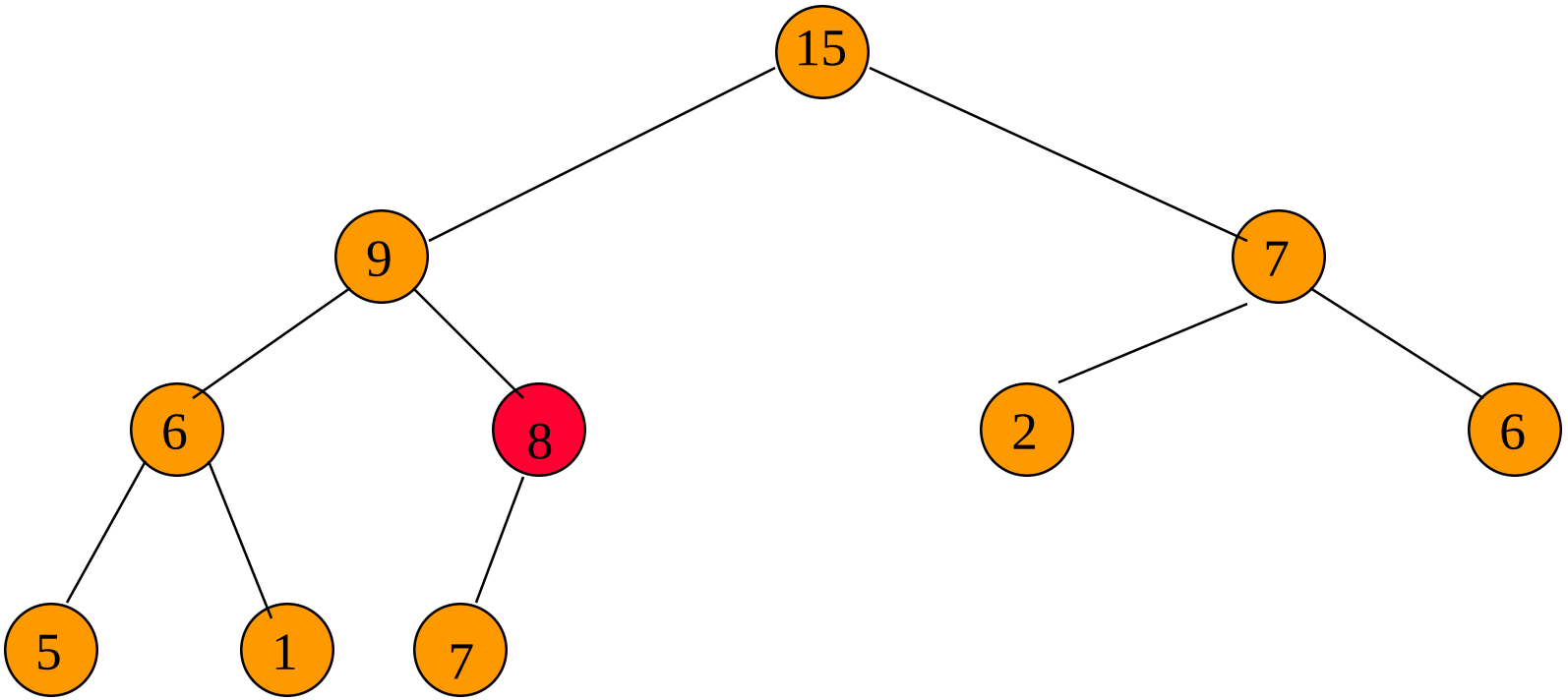

Reinsert 8 into the heap.

# Removing The Max Element



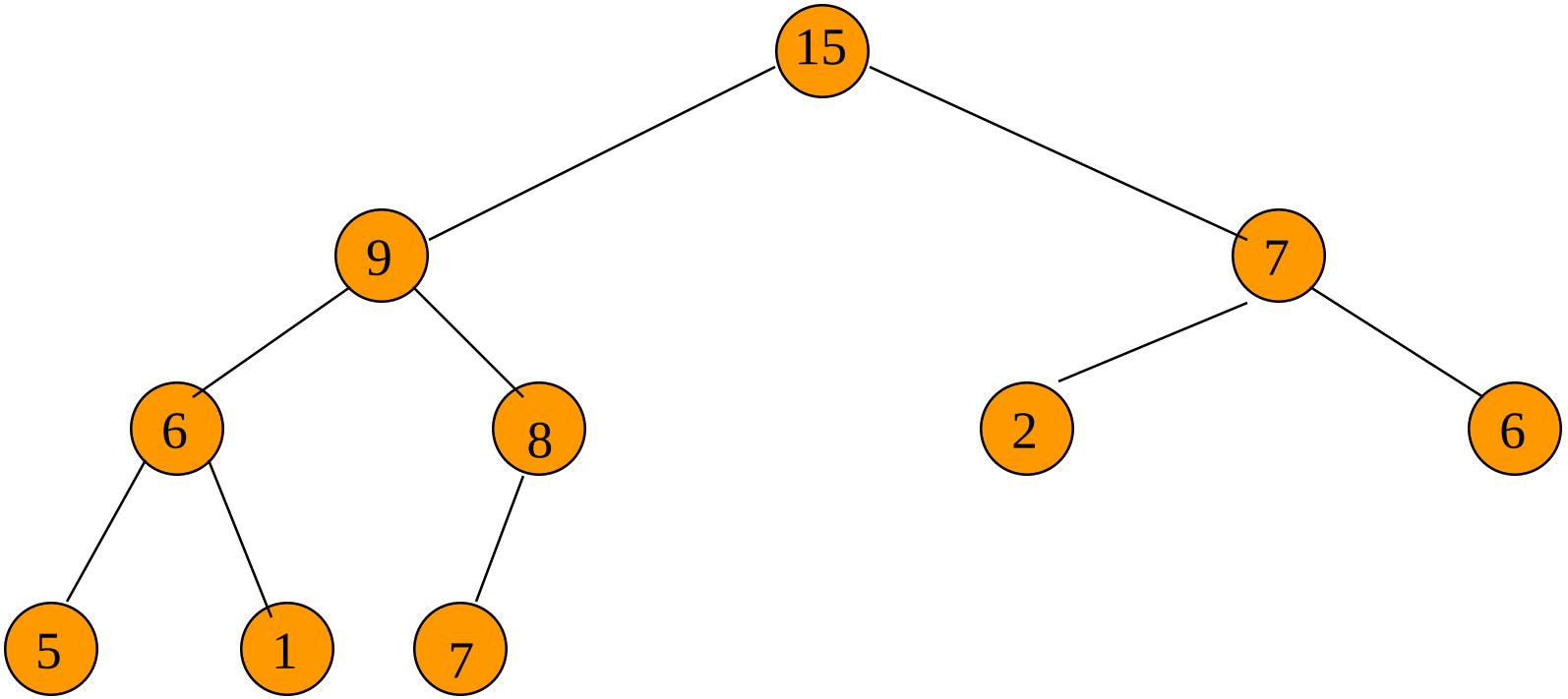

Reinsert 8 into the heap.

# Removing The Max Element



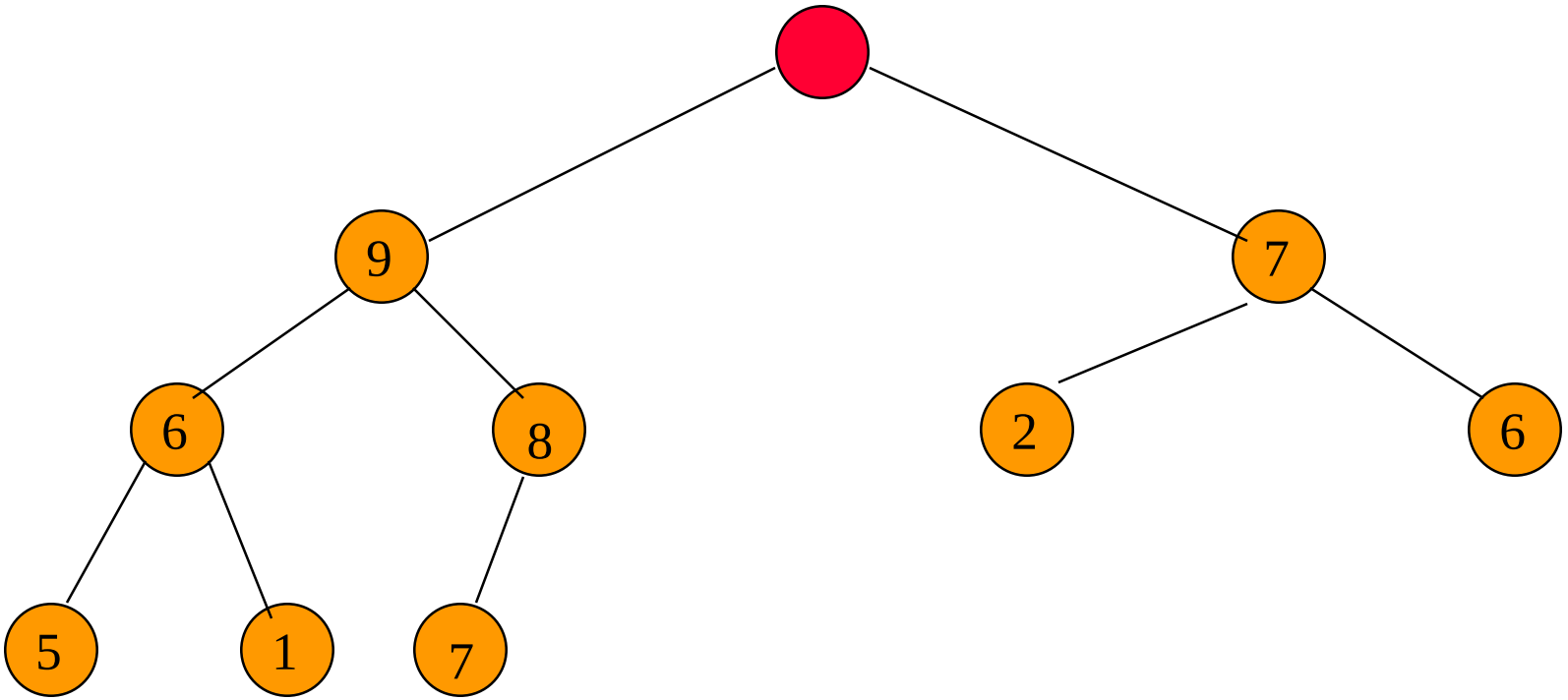

Reinsert 8 into the heap.

# Removing The Max Element



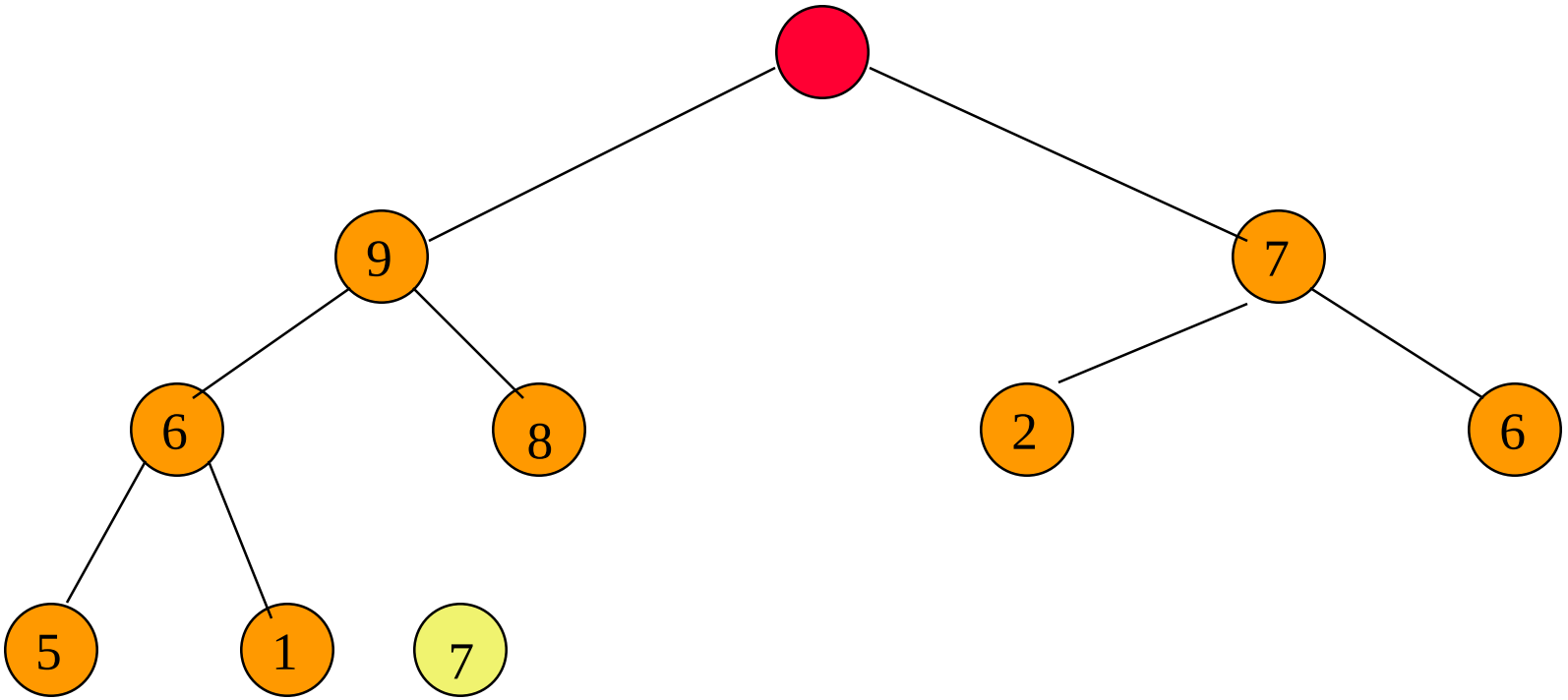

Max element is 15.

# Removing The Max Element



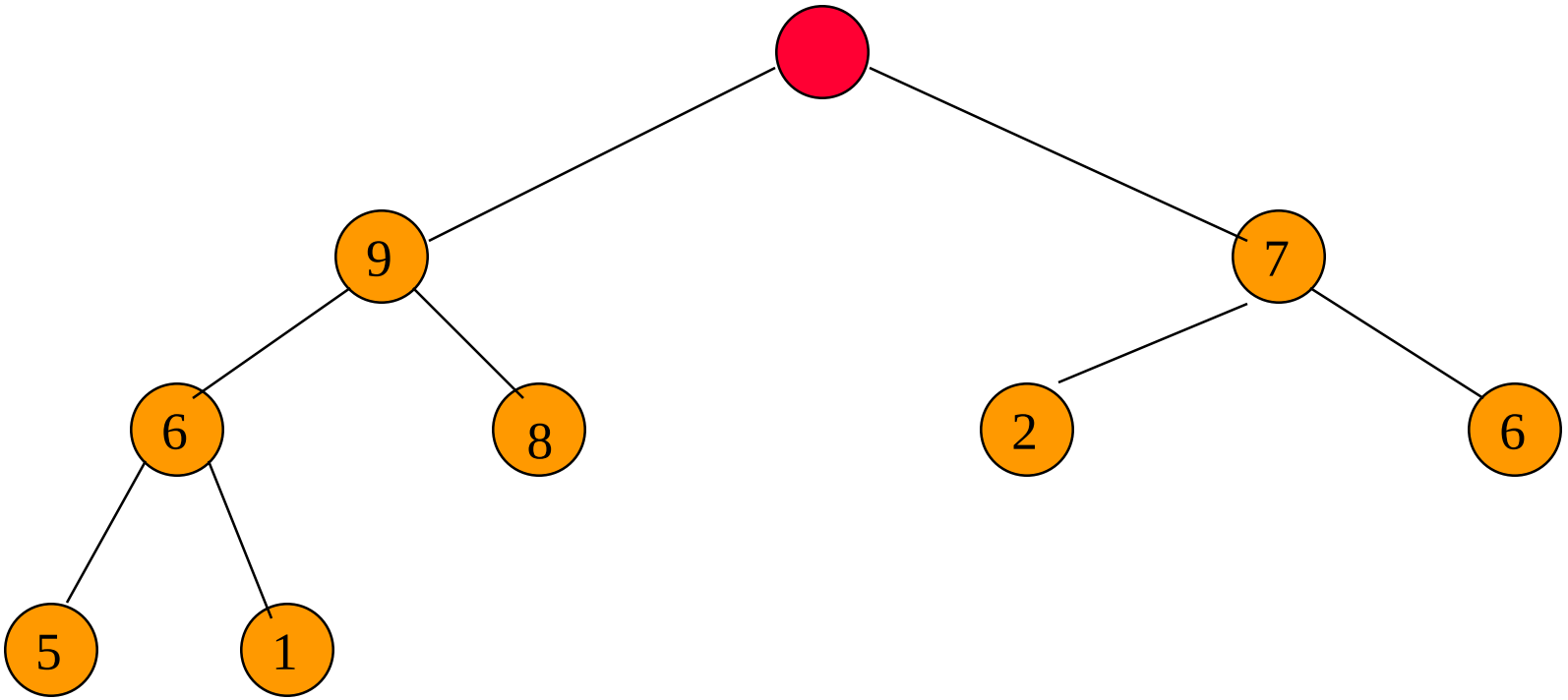

After max element is removed.

# Removing The Max Element



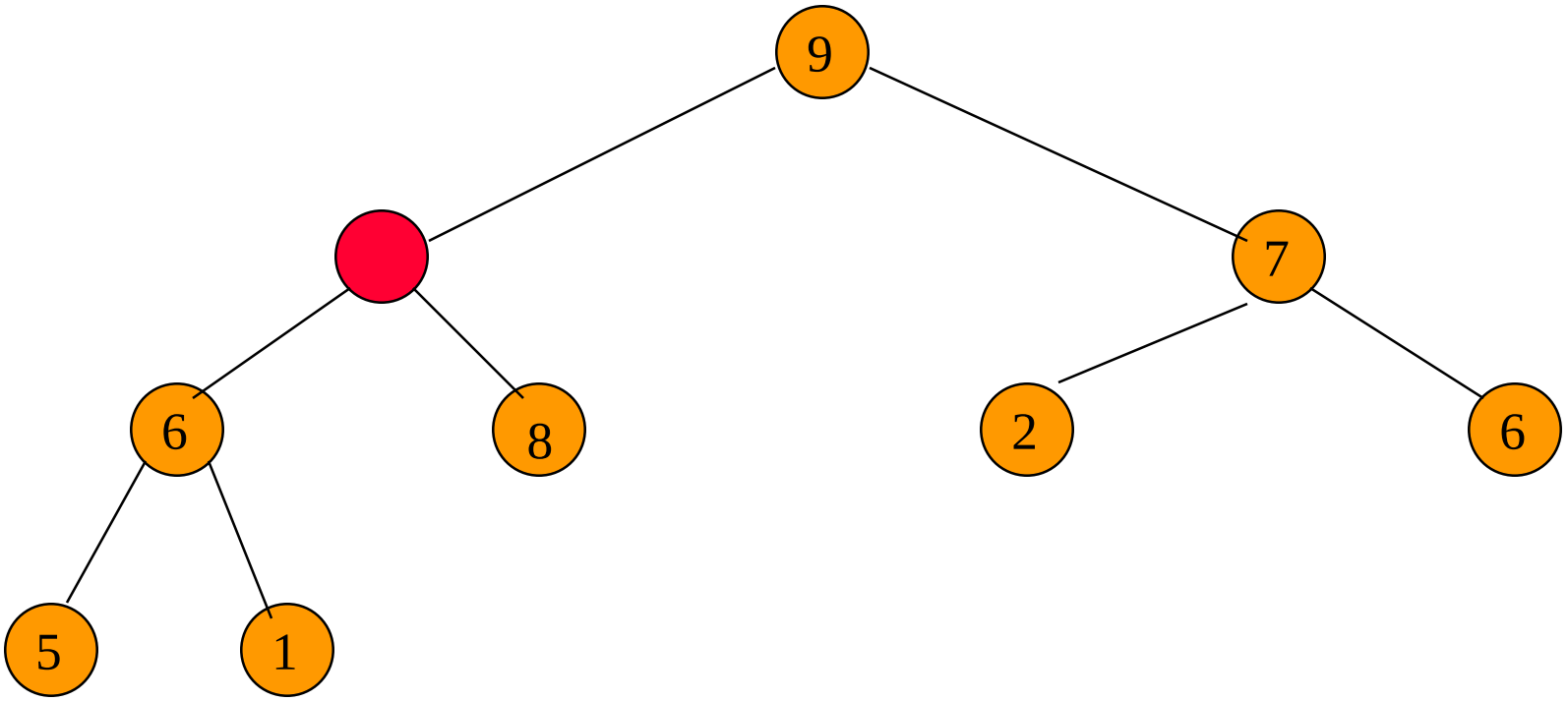

Heap with 9 nodes.

# Removing The Max Element



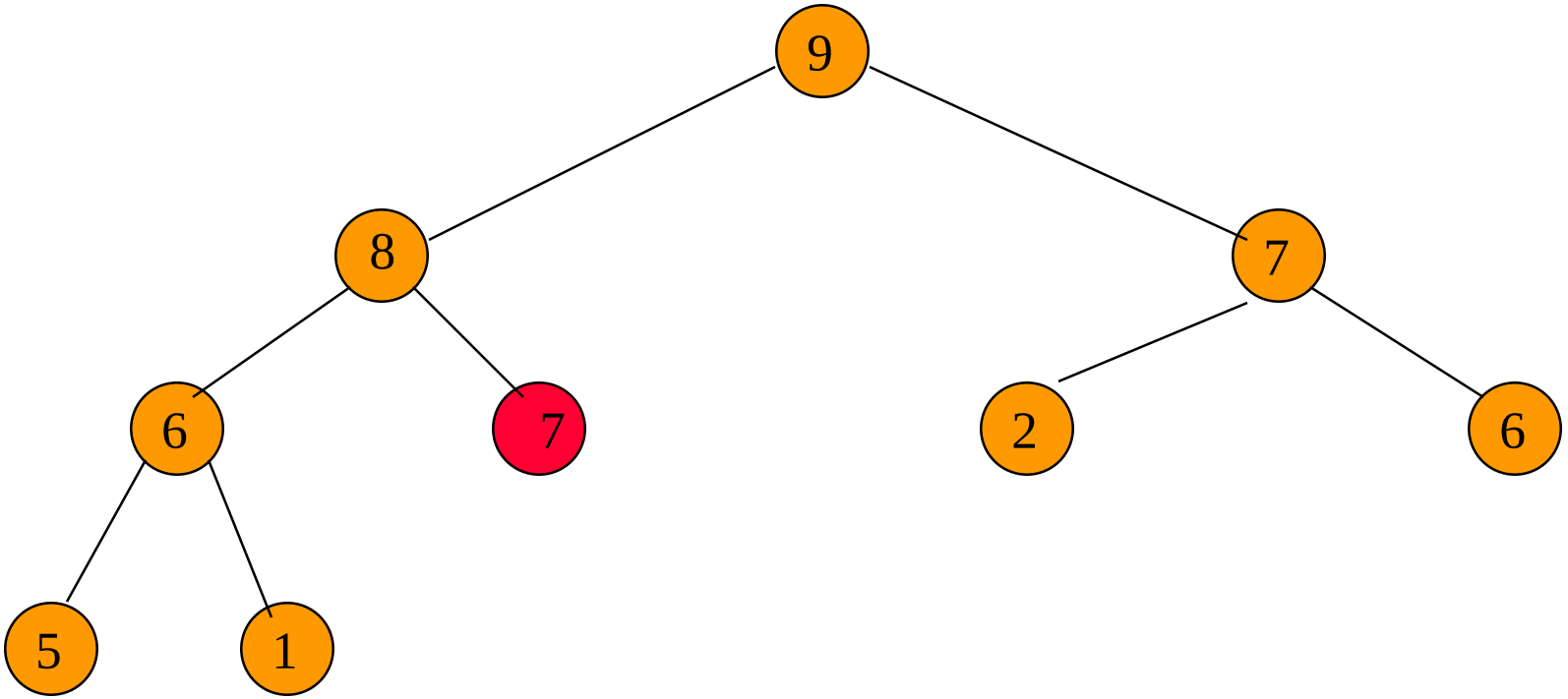

Reinsert 7.

# Removing The Max Element



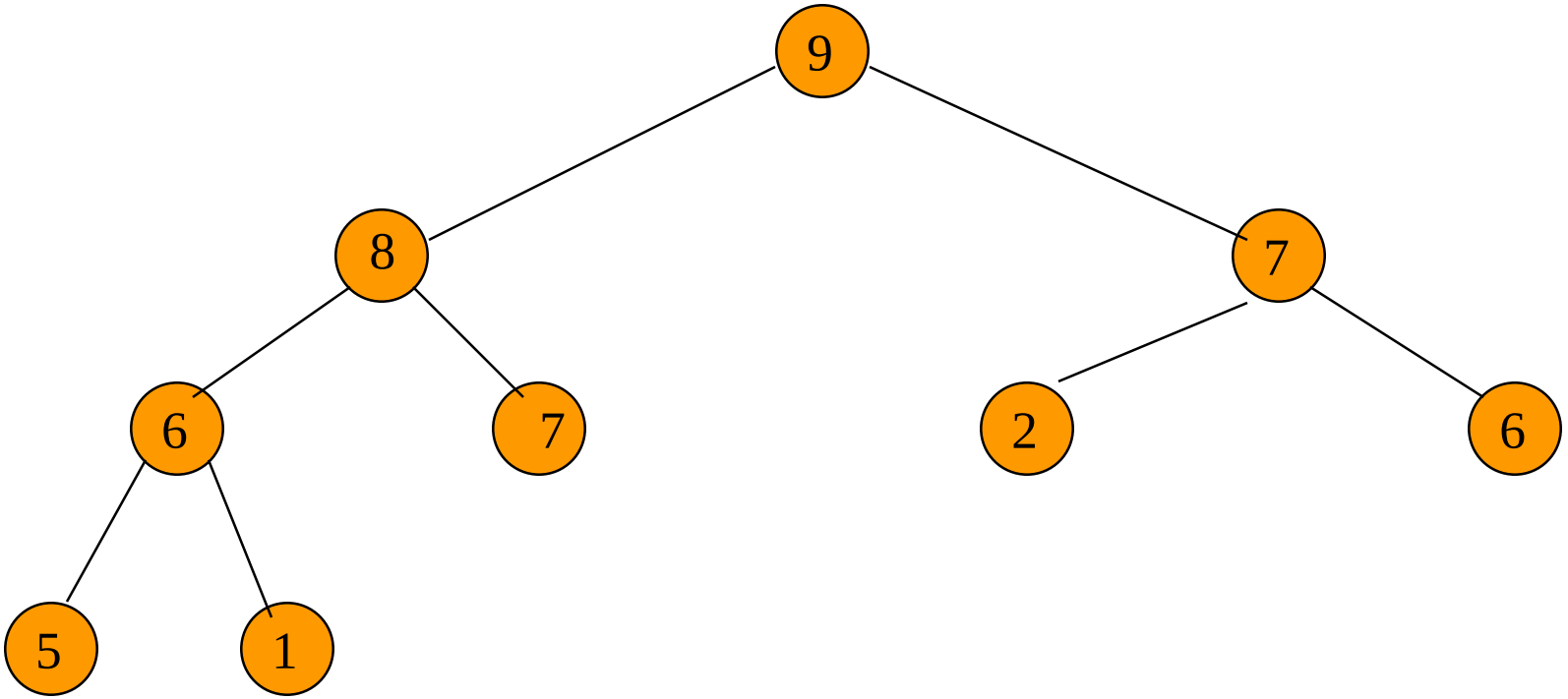

Reinsert 7.

# Removing The Max Element




Reinsert 7.

# Complexity Of Remove Max Element




Complexity is O(log n).